

Spring Boot VPS Deployment Guide

Deploy Your Spring Boot App
on Ubuntu VPS in 15 Minutes
Automatically



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1. Introduction

This manual provides step-by-step instructions to set up an unmanaged Ubuntu VPS server and deploy your Spring Boot website using our setup script. This is an action manual, Read each step then perform instructions immediately. Do not read ahead without executing instructions. This only works if you follow along while performing the task.

1.1 What This Script Does

The script is designed to install and setup software stacks for deploying Java/Jakarta EE Spring Boot website on Ubuntu based servers such as VPS and Cloud servers. The script is non-interactive and can setup the environment with a few clicks. It runs the setup within 10 to 15 minutes to bring your website live.

Installation Stack & Configuration

- Time-zone Configuration
- OS Updates & Upgrades
- Java (JDK)
- MySQL DB Server
- Nginx Server
- Certbot SSL for security
- Firewall for security
- Fail2ban for security
- JAR File Deployment for your web-app

1.2 Who This Is For

- Java/Jakarta EE Developers who built Spring Boot applications but hate VPS set-ups.
- Agencies deploying for clients with Spring Boot apps.
- Hosting resellers who want to setup VPS servers for client tenants (Highly customisable).
- Anyone who knows Java EE but doesn't know how to put their websites to live servers.

1.3 Who This Is NOT For

- Those who want to fix their servers that already deployed.
- Those that have a server that doesn't run on Linux Ubuntu OS.
- Those that are on a Shared hosting cPanel.
- Docker/Kubernetes users.

1.4 Script Requirements

Our script has been tested on the following VPS servers having 1GB RAM & 1 vCPU:

- Hetzner CX22
- DigitalOcean Basic
- Contabo VPS Small

1.4.1. Server Requirements

Components	Minimum	Recommended
CPU	1 vCPU	2 vCPU or upwards
RAM	2GB	4GB or upwards
Disk	20GB	40GB or upwards
Bandwidth	1TB/m	3TB/m or upwards

Table 1.1

1.4.2. OS Requirements

- Ubuntu 24.04 LTS x64 (Recommended)
- Ubuntu 22.04 LTS x64 (Tested, works)
- Debian 12 (Should work, untested)
- Docker (Won't work)

1.4.3. Network/DNS Requirements

Some details should be sent by your hosting provider through email after VPS purchase.

- Root access password or key (Sent by your hosting provider through email)
- VPS Public IPv4 address
- Domain name pointed to VPS Public IPv4 address via A records (Required for SSL)
- Ports 22, 80, 443 open in VPS firewall panel (Most servers already set it up)

1.5 What You'll Get

- None interaction Automated installer: **setup-vps-spboot-A1.sh** handles stack installations, configurations and web file deployment + Let'sEncrypt SSL Certificate setup with auto renewals every 3 months.
- Setup server environment to be Servlet & Web-socket ready.
- Config.env file: Allows you to edit 8 variables instead of 50 lines of config.

2. Prerequisites

2.1. PC Software (Free)

- **PuTTY** (Must have) – download.msi from: <https://putty.org>
- **Notepad++** (Optional but recommended) – download from: <https://notepad-plus-plus.org>
- **Free Password Generator** (Optional but recommended) – download from: <https://apps.microsoft.com> We recommend to download this one: see image below.



Figure 2.1: Free Password Generator

2.2. Hosting Details

Some details should be sent by your hosting provider through email after VPS purchase.

- Domain
- Your Email (Required for SSL)
- Ubuntu VPS
- Server SSH Credentials (Already listed on section 1.4.3.)

3. Preparations

3.1. Install All Required Software's

- Install PuTTY.
- Install Notepad++.
- Install Free Password Generator.

3.2. Move the Extracted Folder to C:\ Drive

You must start by downloading the TSS1.zip file sent to your email and download it to your PC. The zip file contains the main script: **setup-vps-spboot-A1.sh**, **install-config-spboot-A1.env**, This Manual: **SpringBoot-VPS-Deployment-Guide-v1.0.pdf** and **Workbench-Command-TSS1.txt**. If you are reading this Manual, you probably had already downloaded the zip file but If you did not receive the zip file, you can contact us to place an order for setting up your server in no time: coreligics@gmail.com.

You must extract the TSS1.zip file and move the extracted TSS1 folder into the root of your PC Local Drive (C:\). To locate your PC Local Drive (C:\) on Windows: Open Windows Explorer window > On top of window, find the Address Bar > Erase any text in there > Type: C:\ > Then hit ENTER (See Figure: 3.1 below).

WARNING: Do not rename TSS1 folder and it's content files.

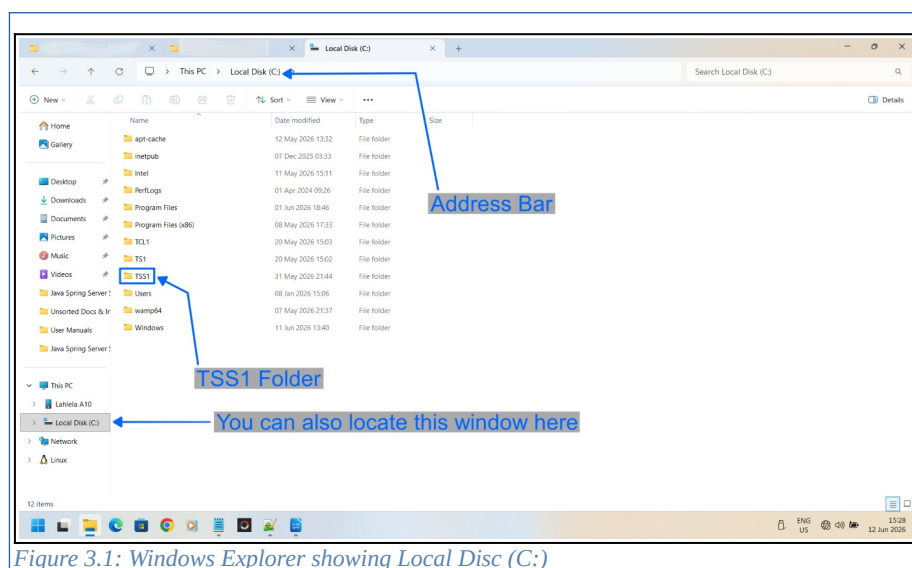


Figure 3.1: Windows Explorer showing Local Disc (C:)

After moving the TSS1 folder into your PC Local Drive (C:\), you must also copy your Spring Boot JAR file into this TSS1 folder if you are planning to deploy it.

3.3. Fill in Workbench Command File

- A Workbench Command File contains command prompts that you are going to use for preparation tasks just by copy and paste into Command Terminal then execute.

- Inside TSS1 folder > Open “Workbench-Commands-TSS1.txt” in Notepad.

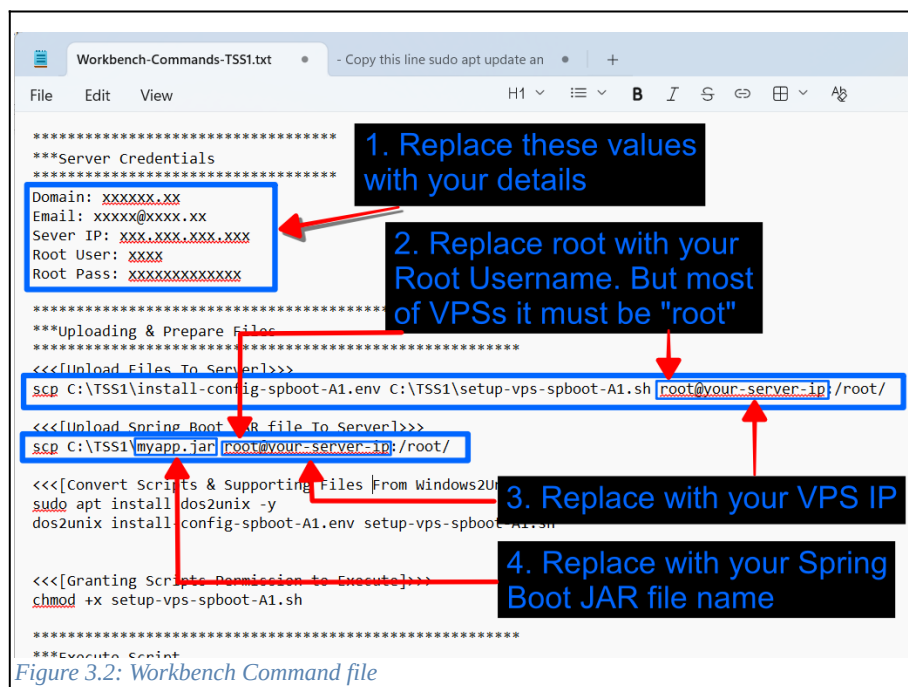
- Under the heading “**Server Credentials**” edit and replace the following values with your details.

Your details should be sent by your hosting provider through email after VPS purchase.

- **Domain:** xxxxxx.xx (Domain that's pointing to your VPS Public IPv4 address)
- **Email:** xxxxx@xxxx.xx (Required by SSL)
- **Sever IP:** xxx.xxx.xxx.xxx (Public IPv4 address)
- **Root User:** xxxxx (SSH Username)
- **Root Pass:** xxxxxxxxxxxxxx (Server Password)

- Under the heading “**Uploading & Prepare Files**” edit and replace the following values with your details:

- myapp.jar
- root@your-server-ip (You can leave “root” as is because most new Ubuntu VPSs user-names are “root”)
- See Fig: 3.2 below for guidance and follow image guidance in order 1>2>3>4.



- Save the Workbench Command File but don't close the Notepad: Click **File** > **Save**.

3.4. Fill in Config File

A Config File (**install-config-spboot-A1.env**) contains a set of global variables that are required to setup the server. So, handle with care.

- Inside **TSS1** folder > Open “**install-config-spboot-A1.env**” in Notepad++.
- In Notepad++ top-bar menu, click: **Language** > **S** > **Shell**. Don’t close the Notepad++ yet.
- Open the “**Free Password Generator**” software and see Figure: 3.3 below on how to configure and generate the proper Database root password with special characters that won’t conflict with the Bash code (Follow image guidance in order 1-8.). Safe characters: **_ - . @**.

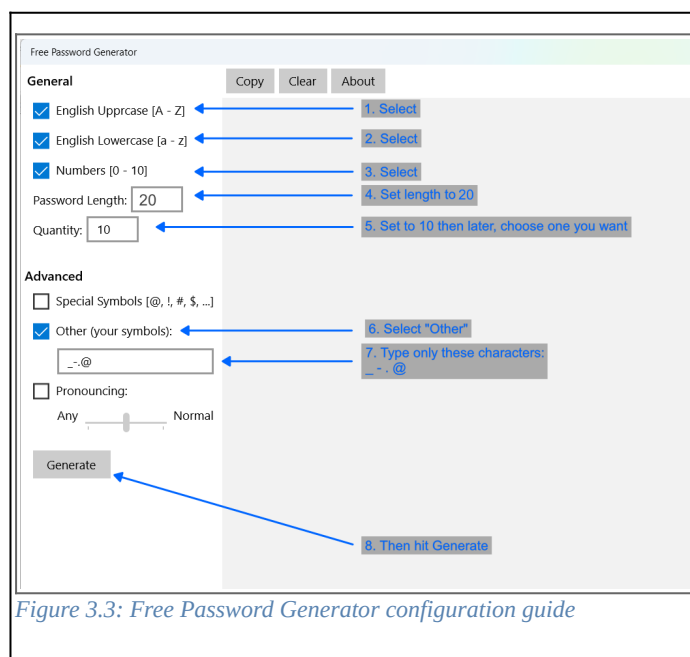


Figure 3.3: Free Password Generator configuration guide

After generating the password, you can leave the program opened (You gonna copy one of these passwords later). Get back to **install-config-spboot-A1.env** file

- Inside the **install-config-spboot-A1.env** edit and replace the following values with your details:
Warning: You must ensure that you don’t include space between characters, or around words and between words. See guidance below.

NOTES: “Required” means: **You must replace default with your value.**

Under The Header 1: “Required Configuration”

Key	Default Value	Action	Description
DOMAIN	“your-domain.com”	(Required)	Your VPS Domain
EMAIL	“admin@mail.com”	(Required)	you@gmail.com

Table 3.1

Under The Header 2: “Application Configuration”

Key	Default Value	Action	Description
JAR_FILE_NAME	“myapp.jar”	(Required) But if you aren’t deploying, leave this default value as is to avoid “variable not set error”.	Your Spring Boot file name.
APP_PORT	“8081”	Don’t change this unless you know what you’re doing.	Your website port. 8081 is recommend to avoid clash with Nginx on port 8080.
SERVER_ADDRESS	“127.0.0.1”	Don’t change this unless you know what you’re doing.	This is a Localhost IP for internal requests between stacks inside server.
TIME_ZONE	“America/New_York”	(Required)	See format & examples in config file comments.

Table 3.2

Under The Header 3: “Database Configuration”

Some of this values **must** match your Spring Boot `configuration.properties` file if you are deploying. If you are not deploying your Spring Boot app, leave the values as default/create your custom values and later when you are ready to deploy, you can change/manage the values using DB UIs listed at the end of this manual or through command lines.

Key	Default Value	Action	Description
DB_ROOT_PASS	“ChangeMeRoot123_”	(Required) Password must only include [A-Z][a-z][0-9] [_ -. @], paste a password from Free Password Generator software.	Main password for MySQL server for manual management later.
DB_NAME	“my_db”	(Required) Must match your Configuration Properties File.	Database name for myapp.jar
DB_USER	“dummy”	(Required) Must match your Configuration Properties File.	Database user for myapp.jar
DB_PASS	“dummy”	(Required) Must match your Configuration Properties File.	Database user password for myapp.jar
DB_HOST	“localhost”	Don’t change this unless you know what you’re doing.	This is Localhost IP for internal requests between stacks inside server.
DB_PORT	“3306”	Don’t change this unless you know what you’re doing.	This is Local port for internal requests between stacks inside server.
DB_DDL_AUTO	“validate”	Don’t change this unless you know what you’re doing.	
DB_SHOW_SQL	“false”	Don’t change this unless you know what you’re doing.	

Table 3.3

Under The Header 4: “Java Configuration”

Key	Default Value	Action	Description
JAVA_VERSION	“17”	Don’t change this unless your app requires different version.	JDK version. Recommended: 17 / 21
JAVA_HEAP_SIZE_XMS	“256” (value in MB)	Don’t change this unless you know what you’re doing.	Java memory size
JAVA_HEAP_SIZE_XMX	“512” (value in MB)	Don’t change this unless you know what you’re doing.	Java memory size

Table 3.4

Under The Header 5: “Fail2ban Configuration”

Key	Default Value	Action	Description
FAIL2BAN_MAX_RETRY	5 (int value)	Don’t change this unless you know what you’re doing.	If users/hackers/bots enters wrong password 5 times, it bans their IPs.
FAIL2BAN_BAN_TIME	3600 (int value in sec)	Don’t change this unless you know what you’re doing.	Brute force IPs will be banned for 3600 sec (1hr).

Table 3.5

Under The Header 6: “Logging”

Key	Default Value	Action	Description
LOG_LEVEL	“INFO”	Don’t change this unless you know what you’re doing.	Events logs
LOG_FILE_NAME	“springboot-setup.log”	Don’t change this unless you know what you’re doing.	File name that will be created to output error logs.

Table 3.6

Under The Header 7: “Spring Profile”

Key	Default Value	Action	Description
SPRING_PROFILES_ACTIVE	“production”	Don’t change this unless you know what you’re doing.	Switching between environments.

Table 3.7

Under The Header 8: “Stack Installation Toggles”

This is the VPS stacks required to deploy Spring Boot app. You only set to “no” if your server already installed a certain stack and the script will skip it during the installation.

Key	Default Value	Action	Description
OS_UPDATE_UPGRADE	“yes”	(Required) “yes” to install & “no” for skip install.	Ubuntu OS refresh and installing new updates.
INSTALL_FIREWALL	“yes”	(Required) “yes” to install & “no” for skip install.	Security module
INSTALL_FAIL2BAN	“yes”	(Required) “yes” to install & “no” for skip install.	Security module
INSTALL_JAVA	“yes”	(Required) “yes” to install & “no” for skip install.	JDK
INSTALL_MYSQL	“yes”	(Required) “yes” to install & “no” for skip install.	Database server
INSTALL_NGINX	“yes”	(Required) “yes” to install & “no” for skip install.	Handles incoming HTTP requests on port 8080.
INSTALL_CERTBOT	“yes”	(Required) “yes” to install & “no” for skip install.	Let’sEncrypt SSL & auto renewals.

Table 3.8

Under The Header 9: “Deployment”

This is where you tell the script to deploy your Spring Boot app (WebApp). You only set to “no” if you don’t want to deploy your app, then the script will skip the deployment during the stack installation. **Tip:** If you currently don’t want to deploy, Later when you are ready to deploy, you can set to “yes” and the rest of stacks listed above on (Stack Installation Toggles) can be set to “no” to tell the script to skip them and only run the deployment.

Key	Default Value	Action	Description
DEPLOY_WEB_FILES	“yes”	(Required) “yes” for jar file deployment & “no” for don’t deploy.	Your Spring Boot WebApp

Table 3.9

Under The Header 10: “Script ID”

Key	Default Value	Action	Description
LICENSE_KEY	“Enter_License_Key_Here”	(Required) Copy keys from the store dashboard where you purchased the script / email.	Script ID
PRODUCT_ID	“xxx”	Never change this.	Script ID
DEVELOPER_ID	“xxx”	Never change this.	Script ID

Table 3.10

- Please double check your details before saving the file.

- You must save all changes, to save all: On Notepad++ toolbar menu click File > Save then close the Notepad++.

3.5. Setting-up & Establishing SSH Connection

When establishing a connection from your PC to your VPS server, you must do so securely using the SSH protocol tunnel. In this section, we are going to setup your server + pc to connect through SSH Protocol for maximum security.

3.5.1. Terminal CLI Basics

From this section we are going to execute few command lines through Terminal (CLI) to prepare the execution of the script to run without problems. If you are not familiar with Terminals, no problem, we are going to run few command lines by copying-pasting. In this manual, we are going to use two types of Terminals (Windows Command Prompt & Putty Terminal). See image below.

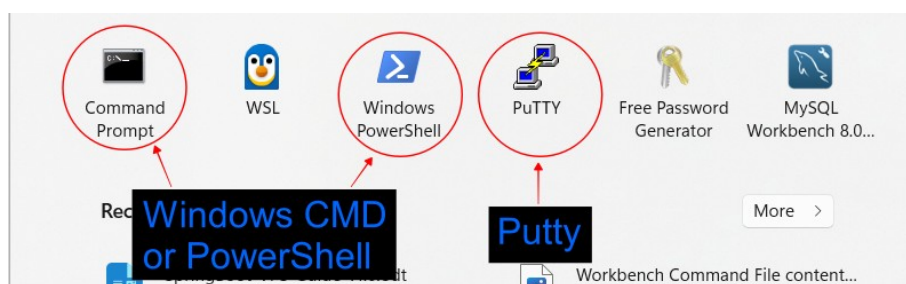


Figure 3.3.1: Showing Windows Terminals (CMD) & Putty Terminal icons

Before we start, you need to understand few things about Terminals:

1. How to copy and paste into the Terminal
2. How to navigate a cursor through a file editor retrieved by a Nano Terminal
3. Terminal behaviour when typing/pasting a password into Terminal.

1. How to copy and paste into the Terminal

To copy a text from any text file on your PC and pasting it into the Terminal using your mouse: you must copy from the text file as usual but pasting it into the Terminal, you just have to Right click once on the Terminal cursor and the text will automatically be pasted into the Terminal. By just Right clicking inside the Terminal cursor, the text will be pasted automatically. Just after copying from the text file, you may Left click once into the Terminal's cursor just to gain back a focus on the cursor, then Right click once to paste.

The keyboard Ctrl+v paste technique doesn't work on Terminals or after right clicking into the Terminal, there won't be any clipboard dialogue that will appear as normal to select paste.

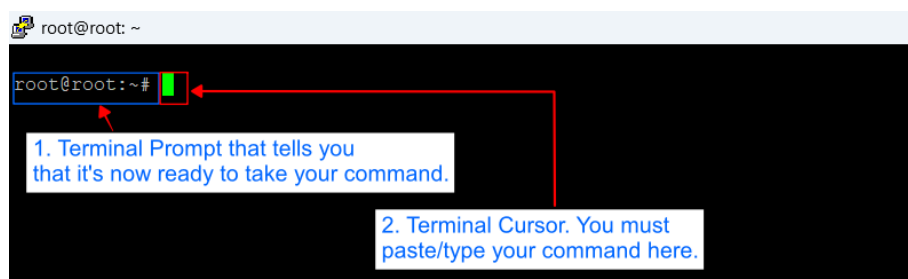


Figure 3.3.2: Terminal Prompt and cursor

2. How to navigate a cursor through file editor in Terminal

If we have some text file from the server that we need to edit through the Terminal, we must retrieve the file by running a command line: `nano filename.txt` and the Terminal will display the file

content so that we can edit it. But to move the Terminal cursor to scroll through the text file, we must use the keyboard navigation arrows instead of using the mouse. The mouse won't move it.

3. Terminal behaviour when typing/pasting a password into Terminal

If a Terminal is asking you to enter your password, the cursor doesn't show progress while typing/pasting the password. It doesn't even show the '*****' or '.....' it just freeze. But while entering the password, the Terminal will be busy accepting the password in the background without showing you the progress and that's a normal security feature for Terminals.

3.5.2. Generating SSH Keys using PuTTYgen

To establish a **secure** connection between your PC to your VPS server, you must connect through an SSH protocol and we must also use SSH keys for authentication rather than using a password. SSH keys are highly secure rather than password. This way we must generate two strong SSH keys using Puttygen software (**Public key + Private key**). By the **Padlock analogy**, the **Public key** will be used as a **Padlock** and must be stored into the server to lock the server. The **Private key** will be used as a male key to unlock the Server (**Server Public key = Padlock**) and must be stored inside your PC.

To generate the SSH keys, follow the instructions below.

- **Opening PuttyGen:** After installing Putty software on your PC, click **Windows Start Button** > **All** > **Putty** > **PuttyGen** the PuTTYgen will open. Or just after clicking the **Windows Start Button**, search for **Puttygen** then open it. Puttygen must come with Putty installation, but if by any reason it's not there. Download it separately from: <https://putty.org> and search for "Puttygen".

- **Generating SSH keys:** See Figure: 3.4 below first. On PuttyGen pane under **Parameters**, select **EdDSA** and the dropdown menu will appear, select value **Ed25519 (255bits)** then hit **Generate** Button. While the program is busy generating the keys, you must wave your mouse repetitively inside the pane until the progress bar finish loading and it will generate 2 keys: **Public Keys & Private Keys**.

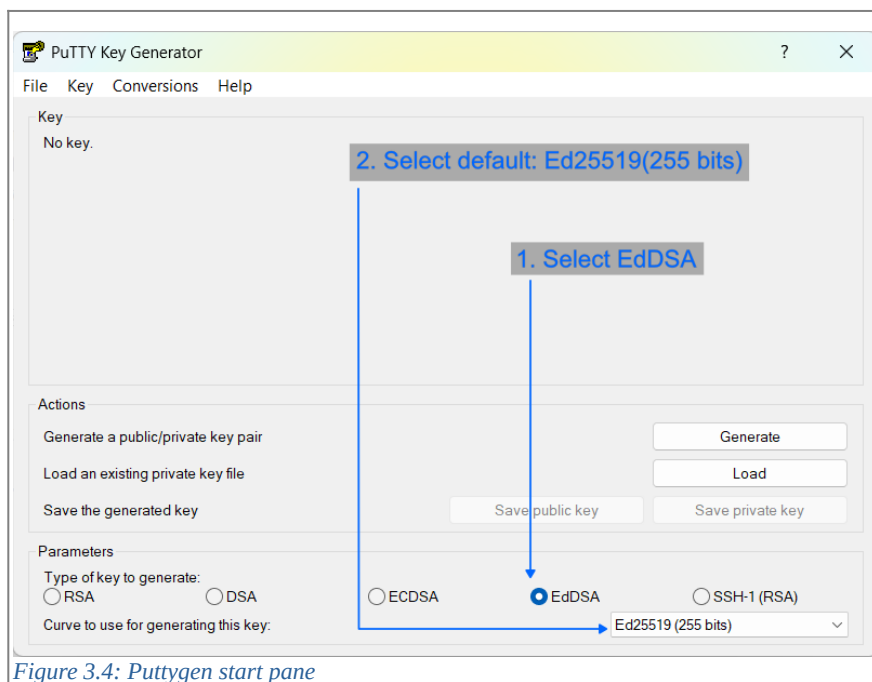


Figure 3.4: Puttygen start pane

- After PuttyGen is finished generating the keys, you can add **Key Pass-phrase** into the pane for extra security but this is optional (*see image below*). If you prefer to enter Passphrase, the Putty Terminal will always ask you to enter your Passphrase every time when you login to your server.

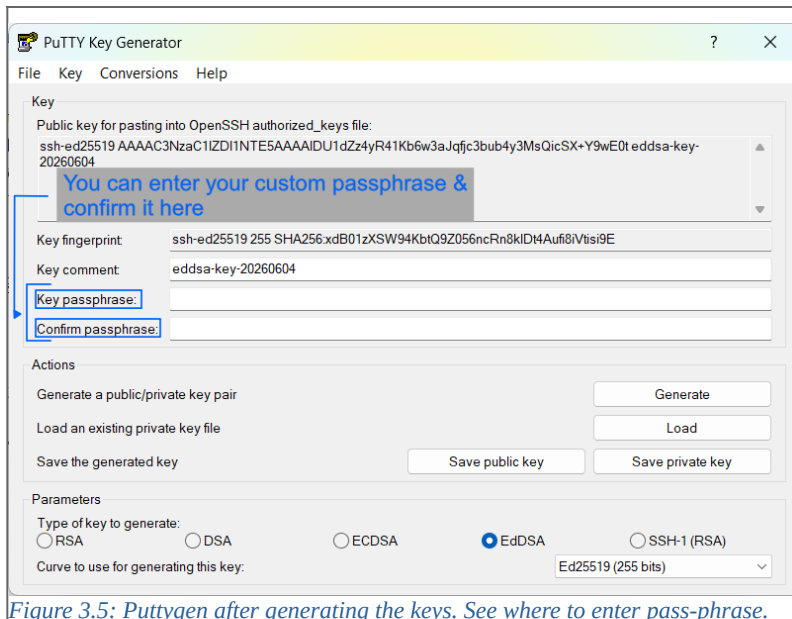


Figure 3.5: Puttygen after generating the keys. See where to enter pass-phrase.

- You must save both keys into your PC before closing the program to prevent loss. Start by saving your **Private key**: Click **Save private key Button** then the window will pop-up, in the window Save as name: type **vps-key-private** and in Save as (**ppk**) file extension then hit Save Button. The key will be stored in **C:\Users\You\.ssh** folder.

- To save the **Public key** before you close the program, highlight and copy the entire **Public Keys** displayed ontop of the pane under the heading: **Public key for pasting...** (See below image first and follow the image guides in order 1-2). then paste them into the **Workbench Commands file** at the bottom, we will need these keys just now.

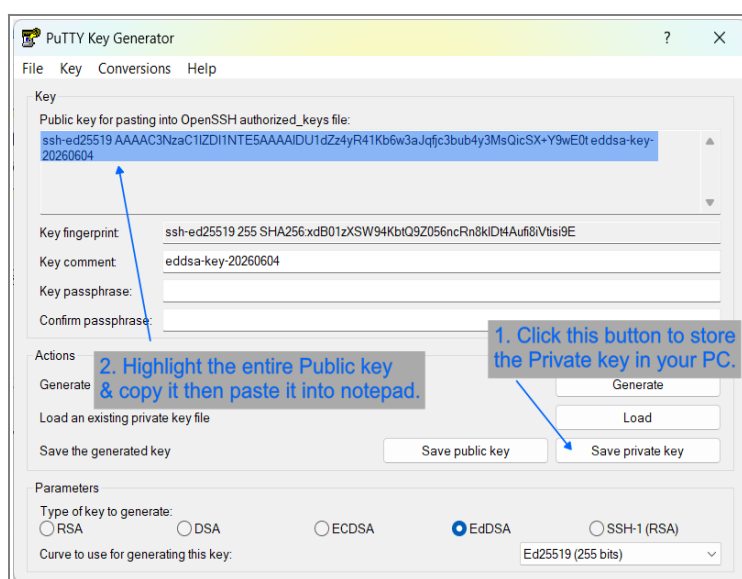


Figure 3.6: Highlight and copy the entire Public key.

3.5.3. Add SSH Public Key to Server

In this section, we must upload your Public Key to your VPS server to lock your server (Public Key = Padlock). There are two options to add the Public Keys to Server.

Option 1: Setup SSH Keys using your Hosting Dashboard

Every hosting company has a different dashboard, it will be impossible for me to give you a **One Instruction to Fit All**, but they all use the same term: “SSH Key” or “Public Key”.

To help you nail this once, here is a setup flow for adding your **Public key** through the Dashboard. Many people make mistakes and miss some steps when they are performing this because it's little bit tricky that's why they fail to do this right. So to give you a little bit of an idea, this Key-Setup-Flow will help you to understand the tricky part so that you can do it right. Just go through this flow and follow instructions after.

Key-Setup-Flow (This is just to give an idea)

1. You must first ADD the Key your account (This means you just add the key to your account but this does not appoint the key to any VPS server yet because you might have multiple VPSs on this current Hosting provider).
2. You must SELECT the VPS you want to appoint this key to.
3. Appoint the key to the actual VPS you just selected (But the VPS still require you to tell it to use this key).
4. You now must TELL the VPS to use this key.

Follow this instructions: (But first See: **Option: 1.1** below to observe how I do it on my Hosting Provider panel and if you stuck you can come back here to try this instructions)

- Login to your hosting provider dashboard and find **VPS Control Panel Button** then click it to open your **VPS Control Panel**.
- Research all over the panel and look for one of these menu names: **SSH Keys/Public Key** but if you don't see this menu button, look under one of these menus: **Security > SSH Keys** or **Account > SSH Keys** or **Settings > SSH Keys** and if found Click to open.
- On the next page: Click **Add SSH Key** or **Create SSH Key** Button and this will display a form.

1. Fill in the form (Add key to acc):

On **Name/Label**: Type your custom name for the key name, e.g. “**windows-pc**” or any name.

In **Public Key**: Paste the Public Key you copied from **PuTTYgen** then Hit **Save/Submit/Add SSH Button** to save the key.

2. Select VPS you want to appoint key to:

In some servers dashboard, you must also have to navigate to the **List Of VPS** page and display a list of your VPS's on the current Hosting Provider (If you have only one VPS, the dashboard will display one VPS).

3-4. Appoint the key & Tell VPS to use the key:

When your list of VPS's are displayed, find and click: **Manage/Settings Button** under the current VPS listed. On the next page: Click **Settings > SSH Keys > Select the key name that you just added > Use SSH Key**, then you are good.

Option 1.1: How I Add a Public Key on my VPS Control Panel

See how I do it on my Hosting Provider Dashboard. Login to your Hosting Provider and navigate to your **VPS Control Panel** similar to the one in Figure: 3.7.0 and follow instructions below (See images below).

1. On your **VPS Control Panel**, click **SSH Keys** menu. (and this will open the **Key List** page, see next image)

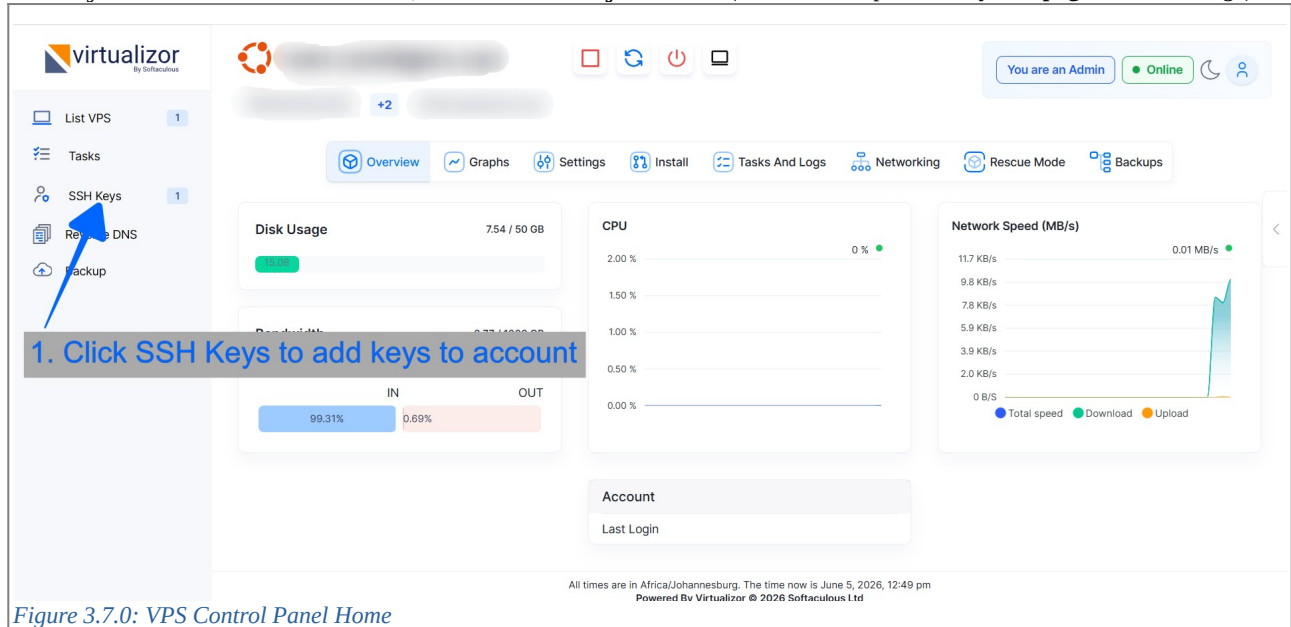


Figure 3.7.0: VPS Control Panel Home

2. On the **Key List** page, click **Add SSH Key** tab. (and this will open the **Add SSH Key** form, see next image)

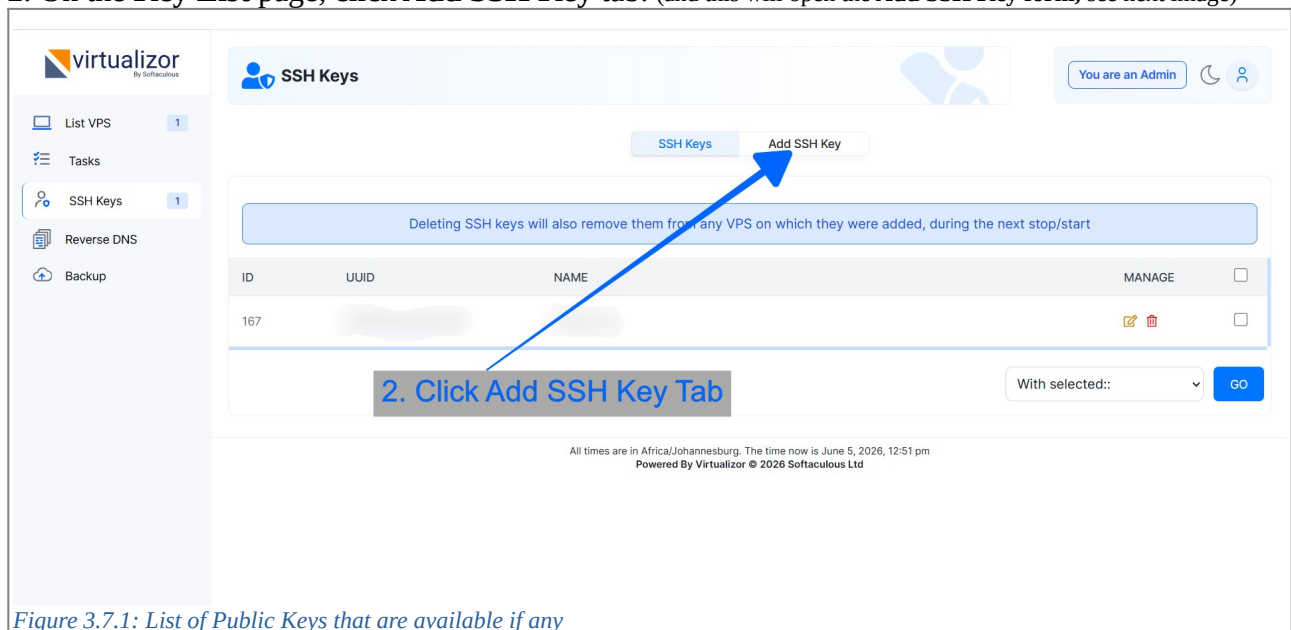


Figure 3.7.1: List of Public Keys that are available if any

3. On **Add SSH Key form**, enter a custom name to name the key: type something like **(windows-pc)** under the **Name** field and under the **Public Key** field paste your **Public key** that you just generated from **Puttygen**. Then click the **Add SSH Key Button** to save. After saving the key, click a **List VPS** tab. Follow the image guides in order 1-4 (and this will open the **List of VPSs**, see next image)

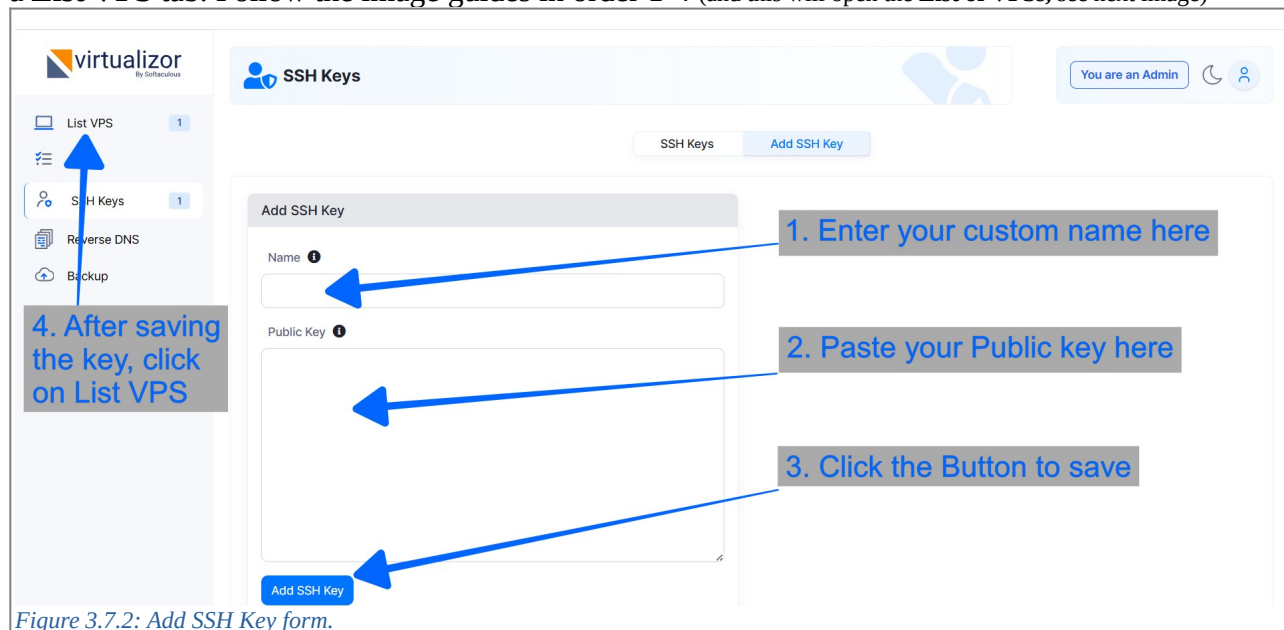


Figure 3.7.2: Add SSH Key form.

4. On **List of VPSs page** you will see all your **VPSs** listed here but if you only have one VPS on this Hosting Provider, you will see only one listed. On your listed **VPS**, click **Manage/Settings Button**. Follow the image guide in order 1-2 (and this will open your **VPS Home screen**, see next image)

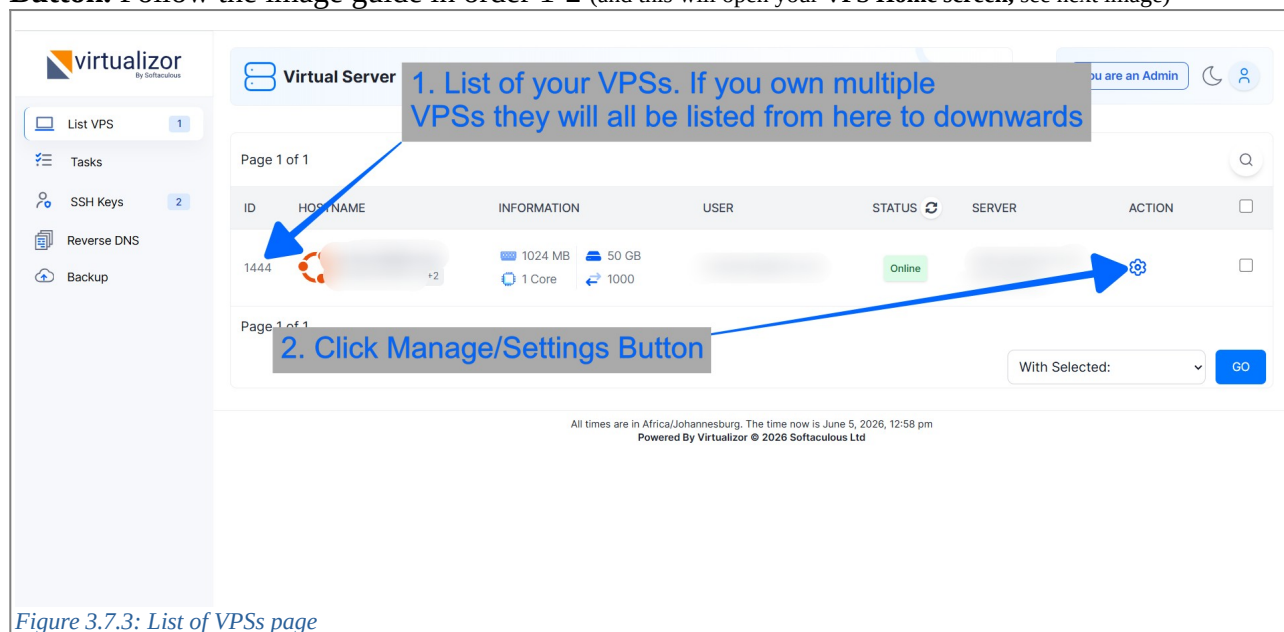


Figure 3.7.3: List of VPSs page

5. On **VPS Home** screen, click **Settings** and to open the **Settings** page.

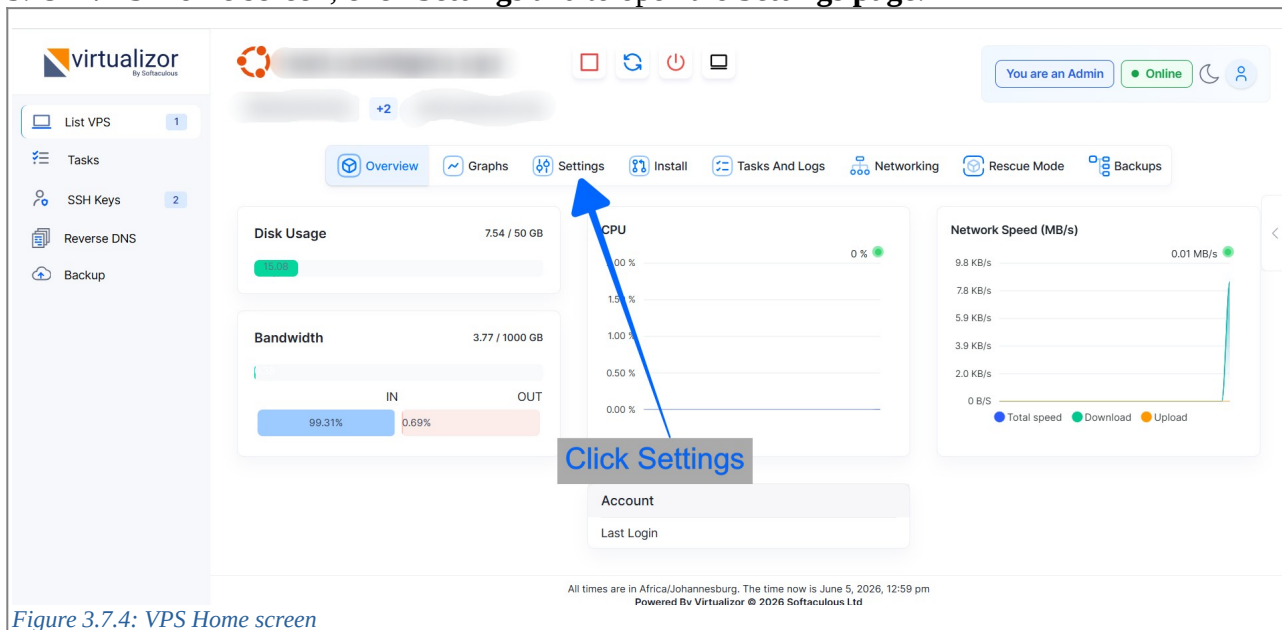


Figure 3.7.4: VPS Home screen

6. On the **Settings** Page, click **SSH Keys** tab to display all **Public** keys you have added to your account panel.

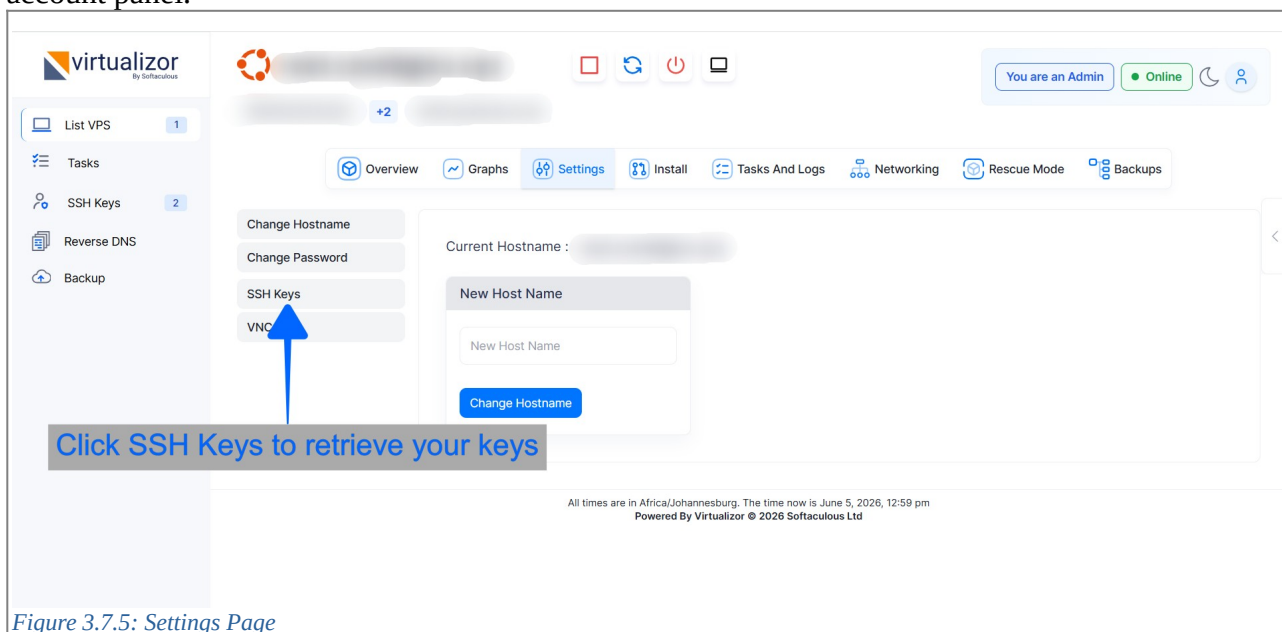


Figure 3.7.5: Settings Page

7. When your keys are displayed, click once on the **key name** to select and click **Use SSH Key Button**. You are done!!!

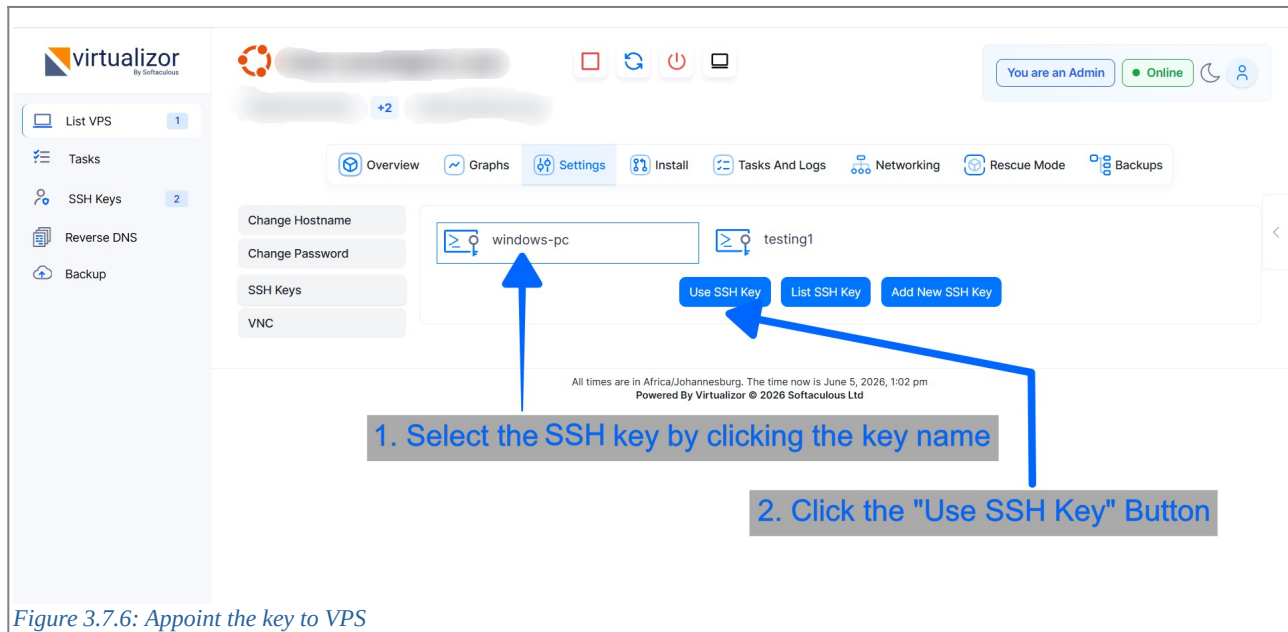


Figure 3.7.6: Appoint the key to VPS

If your Server Dashboard has no “SSH Keys” section, use the next instructions on **Option2** below.

Option2: Setup SSH Keys using Commands

If you successfully added your **SSH Key** using your Hosting Dashboard on **Option1**, you can skip this section.

- After installing Putty on your PC, click **Windows Start Button** > **All** > **Putty** > **Putty** and **Putty** window will open. Or just after clicking the **Windows Start Button**, search for **Putty** then open it.

- **Configuring Putty Terminal:** When Putty opens the window, fill in the following details:

Advice: You can copy and paste your details from Workbench Commands text file.

- **Host Name/IP address:** Paste your server IP address
- **Port:** Leave it on 22 unless you changed SSH port
- **Connection Type:** Select SSH
- **Saved Sessions:** Type a custom name like “**VPSpassword**” to name your Putty settings > On the right side of the window, click **Save Button** (This Button will save your details). See image below first and follow the image guides in order 1-6.
- Make sure your PC is connected to the internet, Then hit **Open Button** at the bottom to connect to your VPS. If it’s your first time connecting with Putty, it will show Security Alert, **Click Accept**.

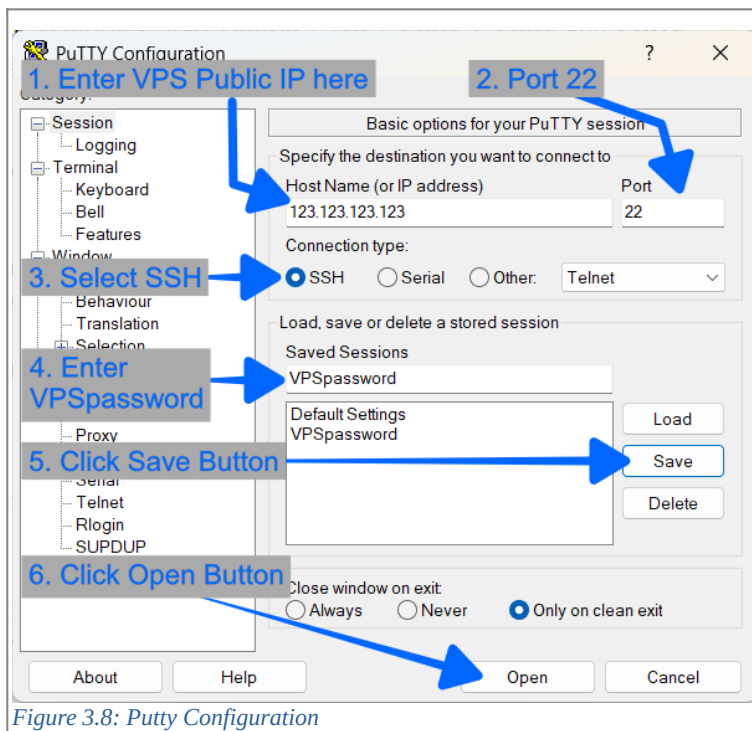


Figure 3.8: Putty Configuration

- The Putty Terminal will show up (Black Screen) asking you: Login as: Type root or your Root Username (If you're not sure about your Root Username, I recommend to type the word: root)> Click Enter. See image below.

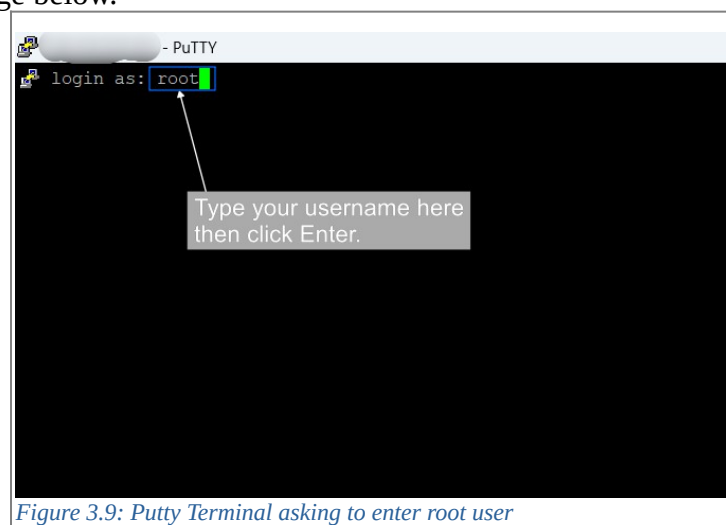


Figure 3.9: Putty Terminal asking to enter root user

- Then it will ask for your server password: Paste/Type your server password (While pasting/typing password, cursor won't move, no stars will show up and no password will show but that's normal) > Click Enter. See image below.

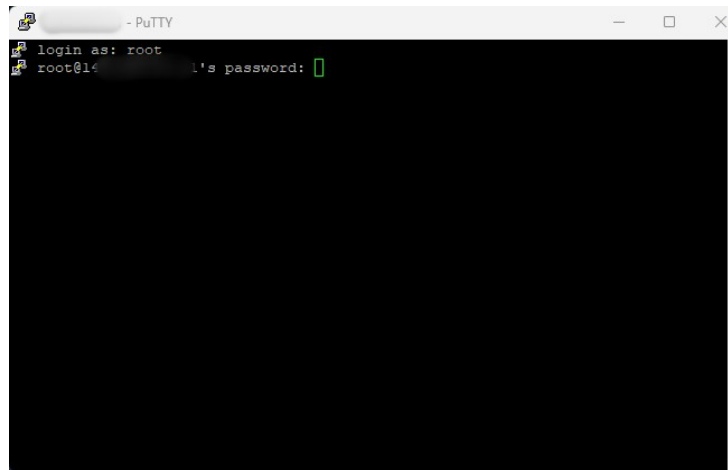


Figure 3.9.1: PuTTY Terminal asking to enter password

- If **password** and **user-name** are correct, the Terminal will display a bunch of info and at the end you will see a prompt similar to this: `root@server:~#` then you are in! See image below.

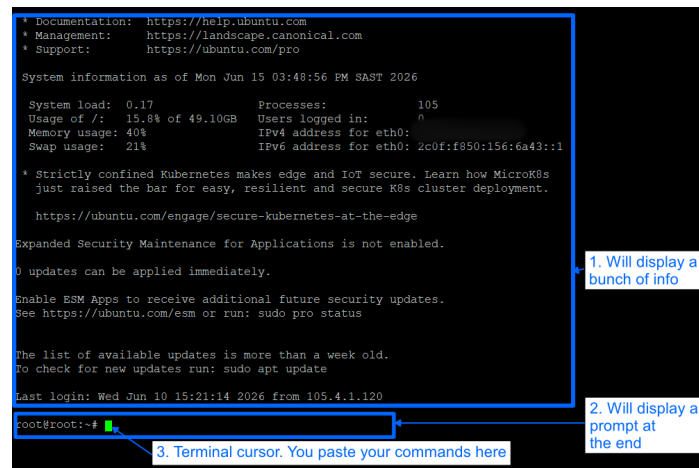


Figure 3.9.2: PuTTY Terminal login success

But if there is an error: **Access denied**, you need to close the PuTTY and re-open it again and try another password or user-name. I recommend you to contact your Hosting Provider and ask them what is your **SSH Credentials** (root user-name + password + IP) then update you Workbench Command file with new credentials.

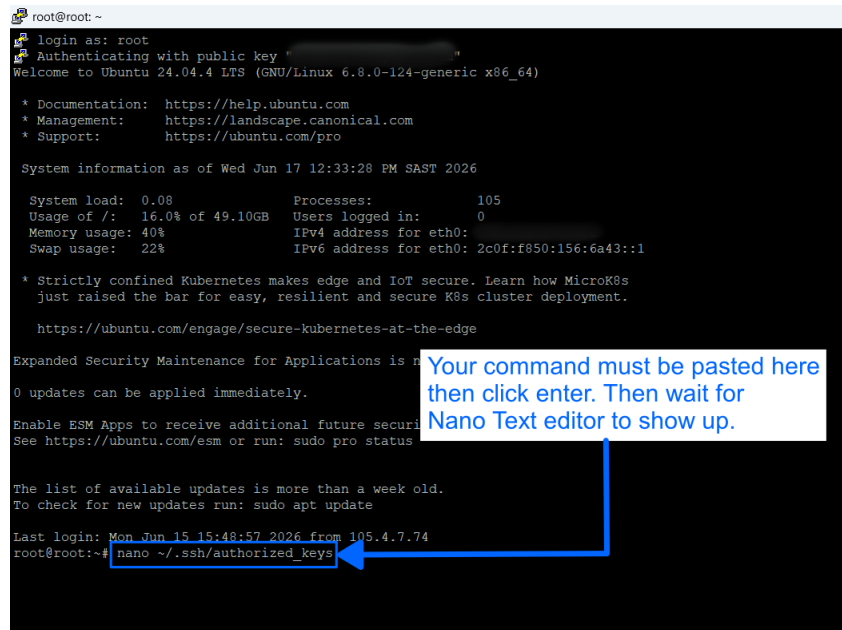
- If your **password** and **user-name** are correct, you will see a prompt similar to this: `root@server:~#`, this means it's done processing and it's now ready to take your next command. This prompt can differ from server to server, it can look like this: `root@server:~#` or `root@root:~#` or `root@your-hostname:~#` or `root@ubuntu:~#` and etc.

- We need to create a SSH folder on your server to hold your `authorized_keys` file if it doesn't exist. Copy this line: `mkdir -p ~/.ssh` and paste it into the Terminal cursor (You don't change anything on this command line) then **click Enter** to execute the command.
- Wait for the Terminal to finish processing until it displays something like this: `root@server:~#`, this means it's done processing and it's now ready to take your next command.
- Let's check whether a SSH folder created successfully. Copy this line: `ls ~/.ssh` and paste it into the Terminal cursor (You don't change anything on this command line) then **click Enter** to execute the command. On results, if you don't see an error, folder created successfully and you can go to the

next step. But if you see an error, go back to the previous step and create the folder by running the command: `mkdir -p ~/.ssh`.

- We need to open your `authorized_keys` file with Nano Text editor so that you can paste your Public key to it. Copy this line: `nano ~/.ssh/authorized_keys` and paste it into the Terminal cursor (You don't change anything on this command line) then click Enter to execute the command. See image below first.

- Wait for the Terminal to open the Nano text editor.



```
root@root: ~
login as: root
Authenticating with public key " "
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-124-generic x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Wed Jun 17 12:33:28 PM SAST 2026

System load:  0.08      Processes:      105
Usage of /:   16.0% of 49.10GB   Users logged in:  0
Memory usage: 40%      IPv4 address for eth0:
Swap usage:   22%         IPv6 address for eth0: 2c0f:f850:156:6a43::1

* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
  just raised the bar for easy, resilient and secure K8s cluster deployment.
  https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not installed
0 updates can be applied immediately.

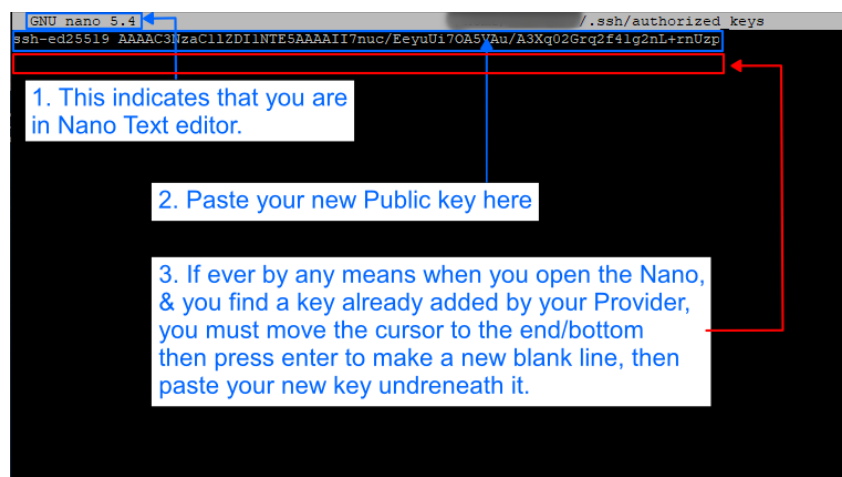
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Mon Jun 15 15:48:57 2026 from 105.4.7.74
root@root:~# nano ~/.ssh/authorized_keys
```

Figure 3.9.3: Calling Nano Text editor to open `authorized_keys` file to be edited

- When the Nano screen appears, Paste the Public Key you copied from PuTTYgen. See image below.



```
GNU nano 5.4 /root/.ssh/authorized_keys
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAII7nuc/EeyuU170A5V7Au/A3Xq02Grq2f41g2nL+rnUzp
```

1. This indicates that you are in Nano Text editor.

2. Paste your new Public key here

3. If ever by any means when you open the Nano, & you find a key already added by your Provider, you must move the cursor to the end/bottom then press enter to make a new blank line, then paste your new key underneath it.

Figure 3.9.4: The Nano Text editor opens the `authorized_keys` file

Now we need to save the key and exit Nano (**Caution:** `ctrl+o` is alphabet: `o`, not a number: `0`). On your keyboard, Click `ctrl+o` to save > then click `Enter` > then click `ctrl+x` to close Nano screen. You should be back to normal prompt: `root@server:~#`

- Copy this line: `chmod 700 ~/.ssh` and paste it into the Terminal cursor (You don't change anything on this command line) then click `Enter` to execute the command.
- Wait for the Terminal to finish processing until it displays something like this: `root@server:~#`, this means it's done processing and it's now ready to take your next command.
- Set permission to the file to restrict access. Copy this line: `chmod 600 ~/.ssh/authorized_keys` and paste it into the Terminal cursor (You don't change anything on this command line) then click `Enter` to execute the command.
- Wait for the Terminal to finish processing until it displays something like this: `root@server:~#`, this means it's done processing and it's now ready to take your next command.
- Close the Putty Terminal.

3.5.4. Test & Verify Your Keys Authentication:

Now that we are done adding your Public key to server, we need to test it and verify whether we configured it correctly. To do this, we need to configure Putty with new settings to be able to use your Private key to unlock the server.

Re-open Putty and fill in the following details:

Advice: You may copy and paste your details from Workbench Commands text file.

- **Host Name/IP address:** Paste your server IP address
- **Port:** Leave it on 22 unless you changed SSH port
- **Connection Type:** Select SSH
- **Saved Sessions:** Type a new name: “VPSkey” to name your Putty settings > On the right side of the window, click **Save Button** (This Button will save your new details for your Private key). See image below first and follow the image guides in order 1-5.

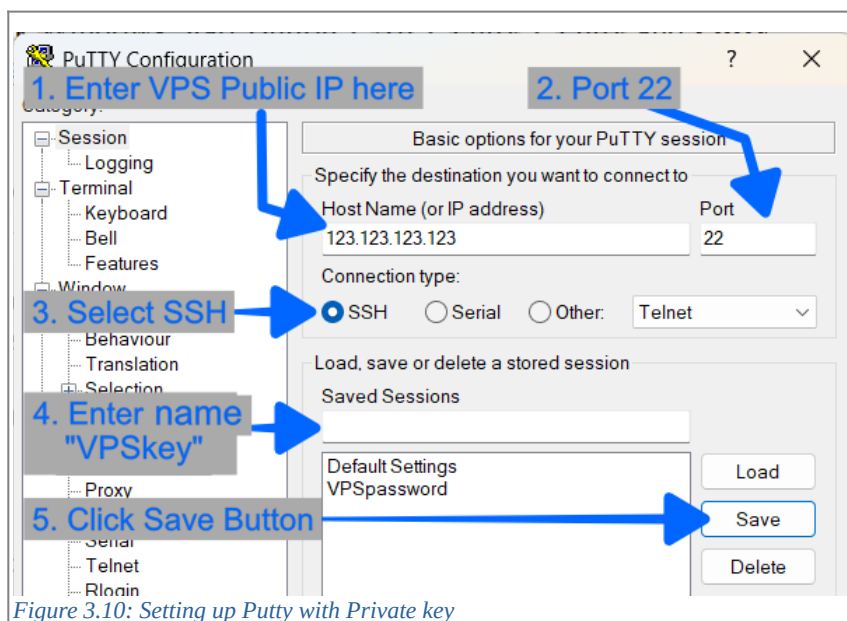


Figure 3.10: Setting up Putty with Private key

- On the left sidebar of Putty window (See Fig: 3.11 below first and follow the image guides in order 1-3): Expand **Session** > Expand **Connection** > Expand **SSH** > Expand **Auth** > click on **Credentials**, on the right, under **Private Key file** for authentication: click **browse button** and browse window will appear. See Fig: 3.11.1 image below.

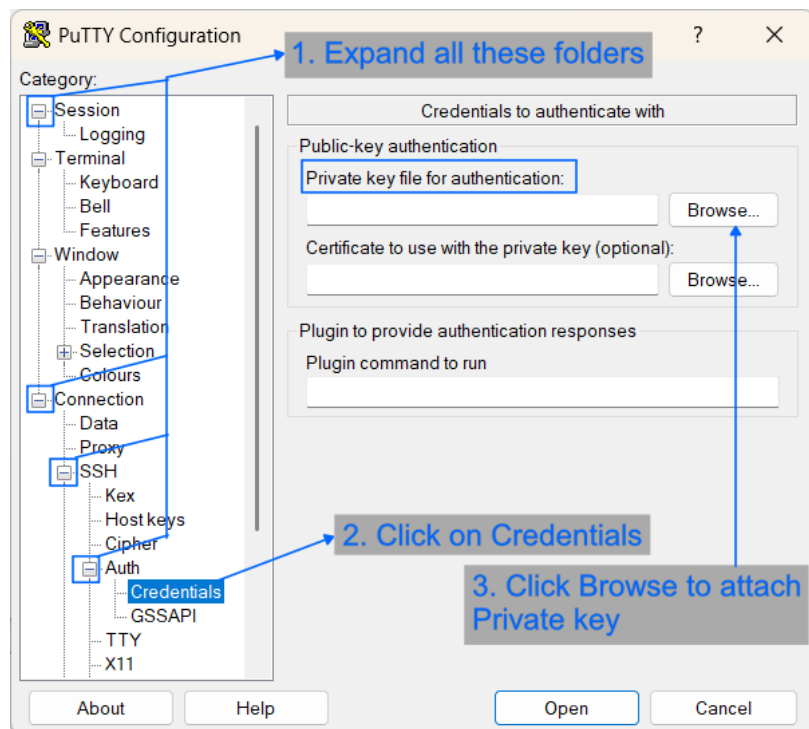


Figure 3.11: Pointing Putty to Private key to use for the connection

When browse window appear, browse in: **C:\Users\You\.ssh** and select the **Private Key file** that you saved earlier e.g. **vps-key-private.ppk** > Click **Open Button** of browse window. See image below first and follow the image guides in order 1-2.

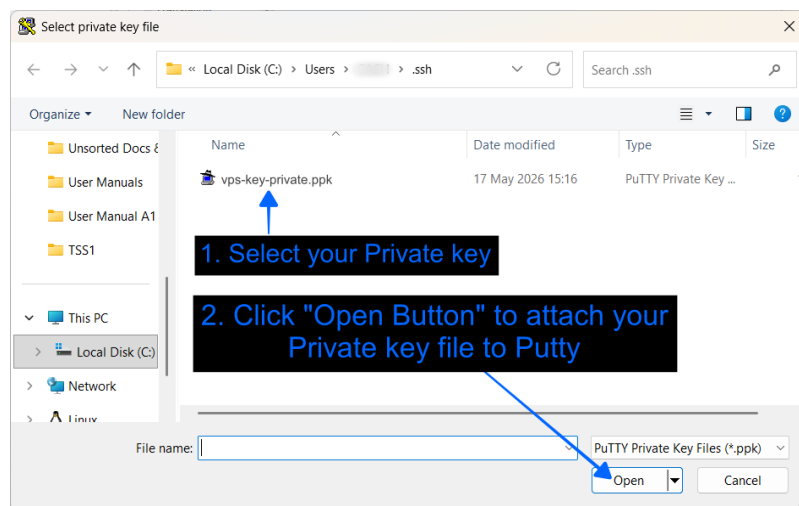


Figure 3.11.1: Attaching Private key to Putty

- On the left sidebar: Click on **Connection** > On **Seconds between keepalives**: Type 60 > Check the box for **Enable TCP keepalives**. Again on the left sidebar: Click **Session** to get back to the start page. See image below and follow the image guides in order 1-4.

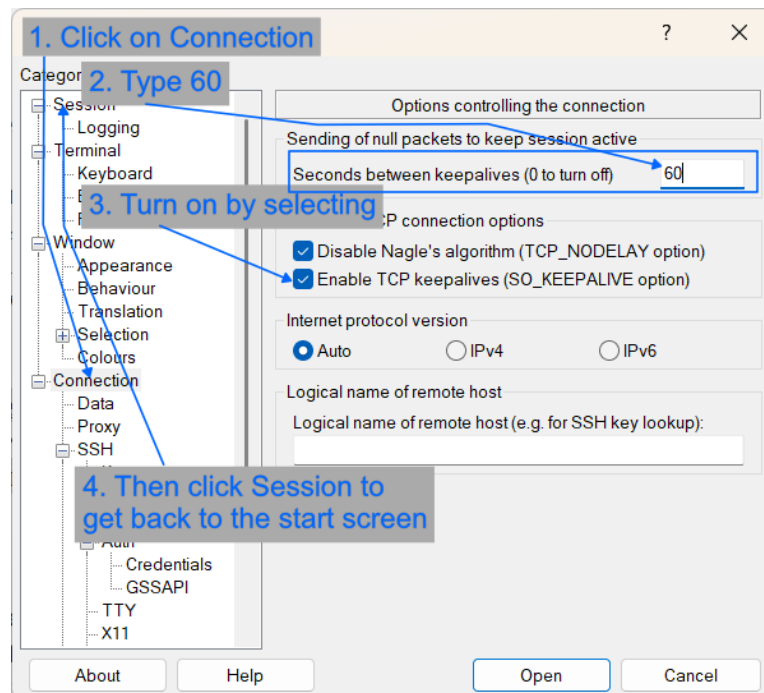


Figure 3.12: Setting up the Keep Alive on Putty

- Click **Save Button** again to save all your new settings (Your new settings will be saved as: **VPSkey**). Make sure your PC is connected to internet, Then hit **Open Button** at the bottom to connect to the server. See image below and follow the image guides in order 1&3.

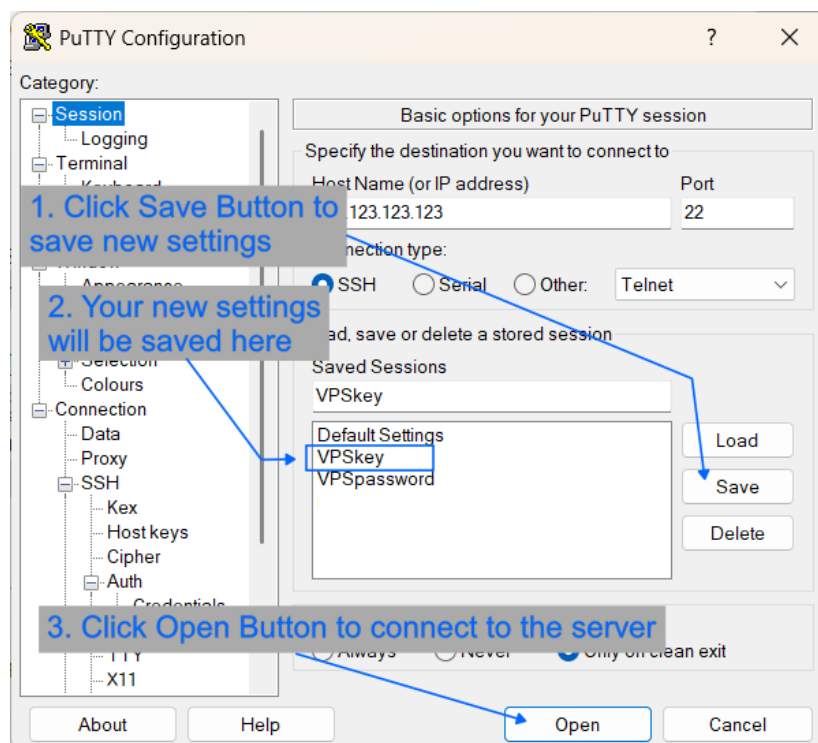


Figure 3.12: Connecting to the Server

Putty Terminal will appear and ask: **Login as:** Type **root** or your **Root Username** (If not sure about your Root Username, I recommend to type “**root**”) > Click **Enter**. See image below.

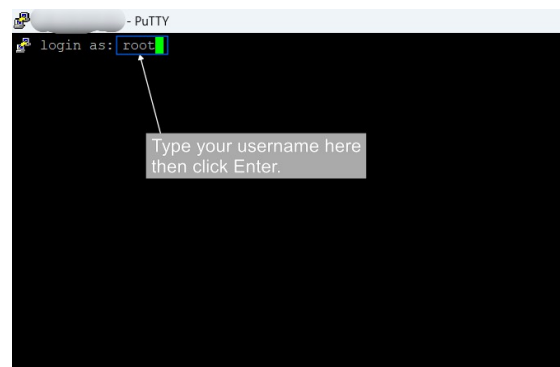


Figure 3.13: Putty Terminal asking for user-name

If key successful: The Putty Terminal will display your info and end with normal prompt: **root@server:~#** without asking for password, this means you successfully connected to your VPS through your Private key. If so, skip fail instruction below.

If key fails: The Putty Terminal will ask for password. Enter password > Click Enter then wait for prompt: **root@server:~#** paste/type this command: **systemctl restart ssh** to restart the keys that failed > then click Enter, close the Putty window and re-open it again. When window opens, under **Saved Sessions:** find and click your saved session name **VPSkey** > Click **Load Button** > **Open Button** to retry connection. See image below and follow the image guides in order 1-3. If you retry the connection for the second time and still ask for password, you must go back to section 3.5.4 and check your details for mistakes then fix them.

Tip

If ever Putty Terminal loses connection with the server, you must close it and reopen it again. Since you’ve already setup it up and saved your connection settings, you don’t have to set it up again. When the window opens again: Navigate to **Saved Sessions:** find and click your saved session name: **VPSkey** > Click **Load Button** > **Open Button** to connect to your VPS. See image below and follow the image guides in order 1-3.

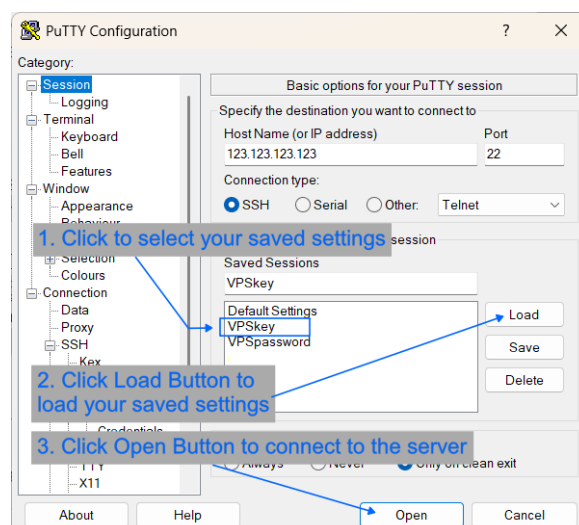
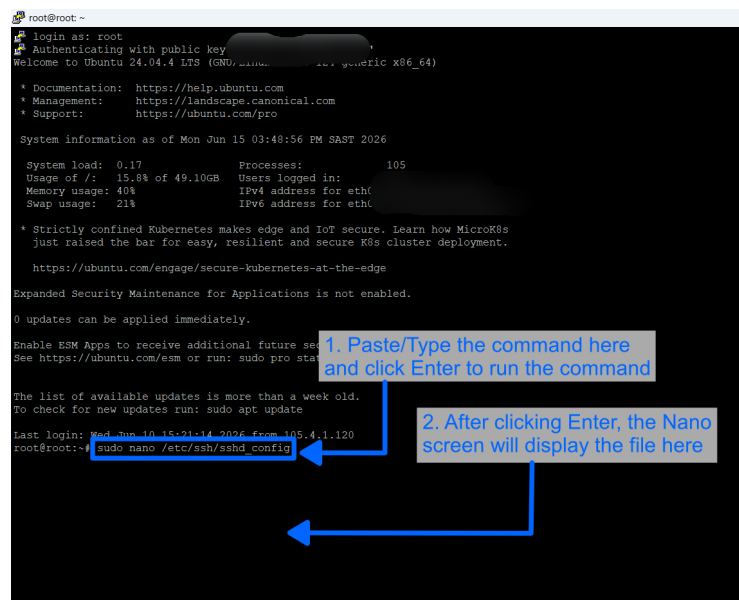


Figure 3.14: Loading your saved settings to connect to the server

3.5.5. Edit SSH Config File

We must edit the **SSH Config** file in the server to prevent the **TimeOut Error** during our script session otherwise the script will stop running in the middle of the session and we will have a mess.

- While the Terminal still connected and showing prompt: `root@server:~#`. Copy this command line: `sudo nano /etc/ssh/sshd_config` and paste it into the Terminal cursor then click **Enter** to execute the command, the Nano Text editor screen will show up displaying the server **SSH Config file** content. See Figure: 3.15 & 3.16 & 3.17 below for clarity.



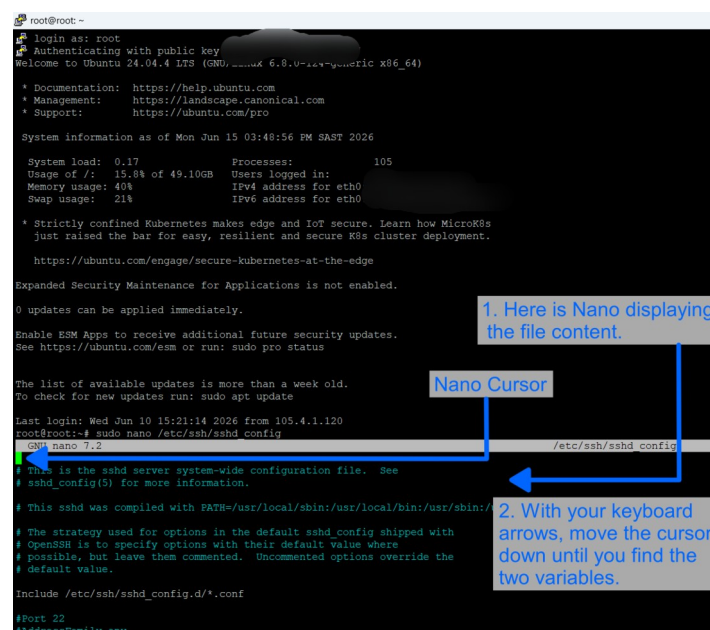
```
root@root: ~  
login as: root  
* Authenticating with public key  
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-100-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
System information as of Mon Jun 15 03:48:56 PM SAST 2026  
  
System load: 0.17          Processes: 105  
Usage of /: 15.8% of 49.10GB Users logged in:  
Memory usage: 40%         IPv4 address for eth0:  
Swap usage: 21%          IPv6 address for eth0:  
  
* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s  
just raised the bar for easy, resilient and secure K8s cluster deployment.  
https://ubuntu.com/engage/secure-kubernetes-at-the-edge  
  
Expanded Security Maintenance for Applications is not enabled.  
  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
  
Last login: Wed Jun 10 15:21:14 2026 from 105.4.1.120  
root@root:~# sudo nano /etc/ssh/sshd_config  
root@root:~#
```

1. Paste/Type the command here and click Enter to run the command

2. After clicking Enter, the Nano screen will display the file here

Figure 3.15: Retrieving the ssh_config file for editing

After running the command to retrieve the ssh config file through Nano, the file will be displayed like this: See image below.



```
root@root: ~  
login as: root  
* Authenticating with public key  
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-100-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
System information as of Mon Jun 15 03:48:56 PM SAST 2026  
  
System load: 0.17          Processes: 105  
Usage of /: 15.8% of 49.10GB Users logged in:  
Memory usage: 40%         IPv4 address for eth0:  
Swap usage: 21%          IPv6 address for eth0:  
  
* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s  
just raised the bar for easy, resilient and secure K8s cluster deployment.  
https://ubuntu.com/engage/secure-kubernetes-at-the-edge  
  
Expanded Security Maintenance for Applications is not enabled.  
  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
  
Last login: Wed Jun 10 15:21:14 2026 from 105.4.1.120  
root@root:~# sudo nano /etc/ssh/sshd config  
root@root:~#  
GNU nano 7.2 /etc/ssh/sshd config  
  
# This is the sshd server system-wide configuration file. See  
# sshd_config(5) for more information.  
  
# This sshd was compiled with PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin  
  
# The strategy used for options in the default sshd config shipped with  
# OpenSSH is to specify options with their default value where  
# possible, but leave them commented. Uncommented options override the  
# default value.  
  
Include /etc/ssh/sshd_config.d/*.conf  
  
#Port 22  
#AddressFamily any
```

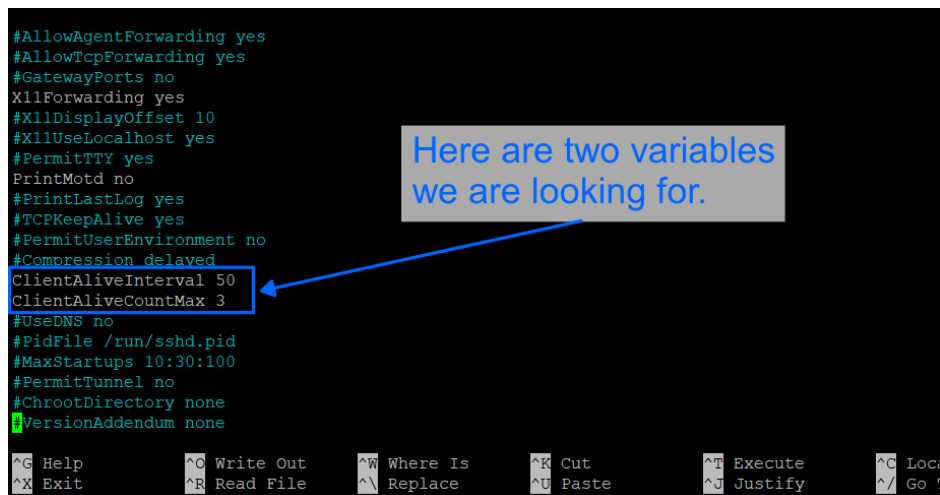
1. Here is Nano displaying the file content.

Nano Cursor

2. With your keyboard arrows, move the cursor down until you find the two variables.

Figure 3.16: Nano displaying the ssh_config file content

You must use your keyboard navigation arrows to move the Nano cursor to scroll down the file and find these two variables: `ClientAliveInterval` and `ClientAliveCountMax`. See image below.



```
#AllowAgentForwarding yes
#AllowTcpForwarding yes
#GatewayPorts no
X11Forwarding yes
#X11DisplayOffset 10
#X11UseLocalhost yes
#PermitTTY yes
PrintMotd no
#PrintLastLog yes
#TCPKeepAlive yes
#PermitUserEnvironment no
#Compression delayed
ClientAliveInterval 50
ClientAliveCountMax 3
#UseDNS no
#PidFile /run/ssh.pid
#MaxStartups 10:30:100
#PermitTunnel no
#ChrootDirectory none
#VersionAddendum none

^G Help    ^O Write Out  ^W Where Is  ^K Cut       ^T Execute
^X Exit    ^R Read File  ^\ Replace   ^U Paste     ^J Justify
^_ Go To
```

Figure 3.17: Keep-alive variables to avoid Time-out error for connection drop-outs.

You must use your keyboard **Backspace Button** to replace the values and set them to: `ClientAliveInterval 50` and `ClientAliveCountMax 3` but if you find your variables already matching our values (50 & 3), but you find them turned off with the signs “#” at the beginning of each variable, remove the signs “#” to activate the variables. The sign “#” in Bash language means comment and this means they turned them off by commenting them out with “#”.

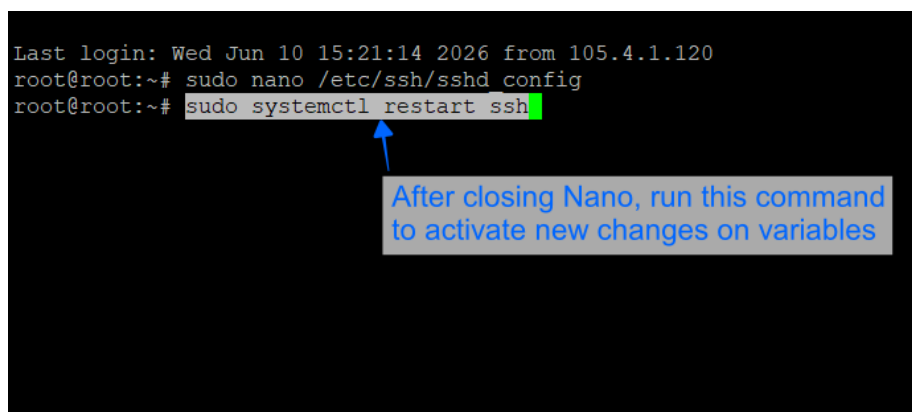
If you don’t find these two variables inside your `ssh_config` file, you must create new ones and make sure to avoid typo mistakes. You can copy and paste: `ClientAliveInterval 50` and `ClientAliveCountMax 3`.

- Now we need to save changes and close the Nano screen (Caution: `ctrl+o` is alphabet: `o`, not a number: `0`): On your keyboard, click `ctrl+o` to save changes > then click `Enter` > then click `ctrl+x` to exit the Nano Screen.

You should be back to the prompt: `root@server:~#`.

- Let’s restart the SSH daemon to update these variables values: Copy this command line: `sudo systemctl restart ssh` and paste it into the Terminal cursor then click `Enter` to execute the command.

- Wait for the Terminal to finish processing until it displays something like this: `root@server:~#`, this means it’s done processing and it’s now ready to take your next command.



```
Last login: Wed Jun 10 15:21:14 2026 from 105.4.1.120
root@root:~# sudo nano /etc/ssh/sshd config
root@root:~# sudo systemctl restart ssh
```

Figure 3.18: Updating the system with new changes

3.5.6. Testing The Domain + IP

Before we run the script, we must make sure that your Domain is pointed correctly to your VPS Public IP otherwise the script gonna fail to install Nginx and SSL Certificate. There are two methods I'm going to show you: **Online method** (Recommended) & the **Command line method** and you must choose the one you like to use.

3.5.6.1. Checking Your Domain Configuration

But first, login to your Domain Provider to check whether your Domain is configured correctly. On your provider dashboard, go to: **Domains** > on **your-domain.com** click **Manage/Setting Button** > **Manage DNS Records**. On **Manage DNS Records** screen, find a section similar to the one in the image below and your Domain settings should be configured similar to this. If you find that your Domain does not point to your **VPS Public IP**, you must click edit to configure it properly to match with the image values. See image below and follow the image guides in order from the top and downwards. Also see the Table below for guidance.

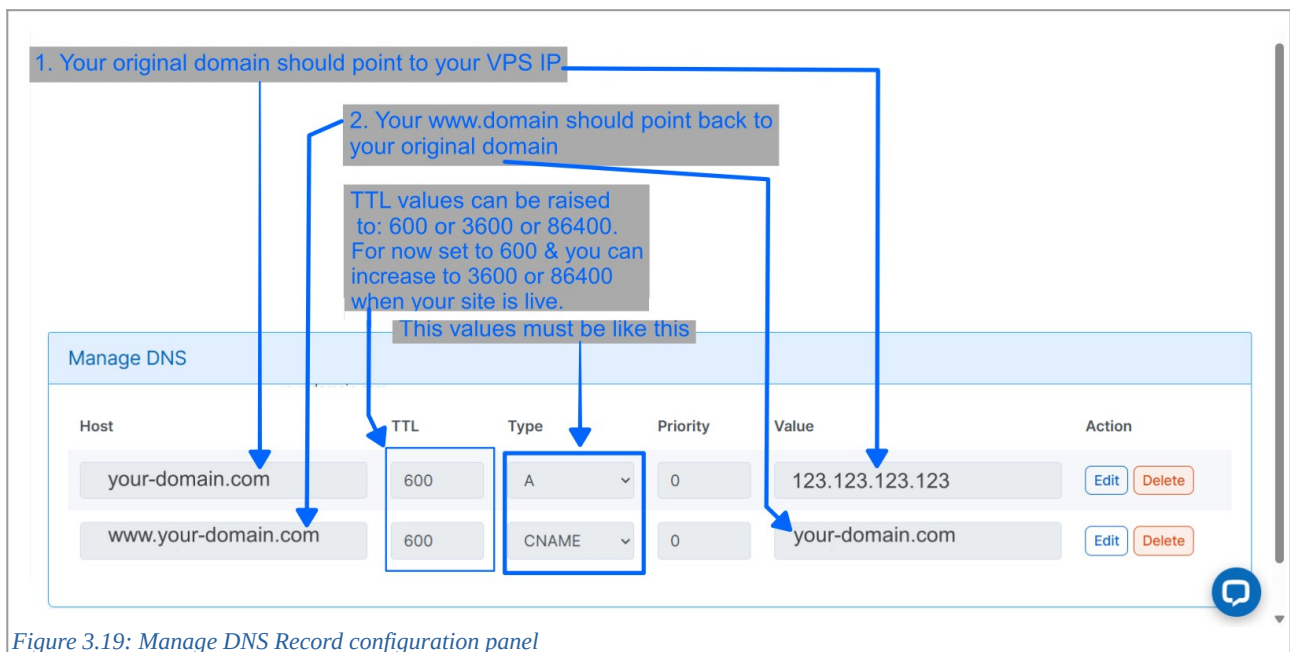


Figure 3.19: Manage DNS Record configuration panel

Host	TTL	Type	Priority	Value
your-domain.com	600	A	0	your-vps-ip
www.your-domain.com	600	CNAME	0	your-domain.com

Table 3.11

If everything looks good, proceed to the following section: **3.5.6.2**. However, you must take into consideration that the Domain propagation can take about 10 minutes to 24 hours to propagate, so after configuring your Domain to point to your IP, keep on testing using methods below until you see your IP appears (Setting TTL to 600 can take about 10 minutes to propagate). Never run the script until you sort out the Domain + IP issue.

3.5.6.2. Testing Your Domain IP & Propagation

Choose one option below to test your Domain IP address:

1. [Option1] Checking Your Domain using Online Tools

Find online tools on Google search and search for [dns checker tools](#) or [what's my dns](#). Test both [yourDomain.com](#) and [www.yourDomain.com](#), select A Record then search. It should return your VPS Public IP otherwise you must fix your domain (See section: 3.5.6.1 above for guidance).

2. [Option2] Checking Your Domain using Commands

- On your PC click Windows Start Button > All > Terminal. Or just after clicking the Windows Start Button, search for `cmd` then open it.
- Copy this command line: `nslookup yourDomain.com` and paste it into the Terminal cursor (You must replace: `yourDomain.com` with your domain) then [click Enter](#) to execute the command. Then wait for results, it should show your VPS IP.
- And also try `www`, Copy this command line: `nslookup www.yourDomain.com` and paste it into the Terminal cursor (You must replace: `yourDomain.com` with your domain) then [click Enter](#) to execute the command. and wait for results, it should show your VPS IP.
- Otherwise you should fix it on your Domain provider before running the script (See section: 3.5.6.1 above for guidance).

4. Server Set-up

Since from the beginning of this manual we were busy preparing everything to be in order for our main script to run without problems. Now we are in a section where we are ready to run our script to setup your VPS server and deploy your website. Please carefully follow these steps below to start setting up your server.

4.1. Upload Files to Server

[STEP 1] Open Windows Terminal:

- On your PC click Windows Start Button > All > Windows Tools > Command Prompt then open it. Or click Windows Start Button > All > Terminal then open it. Or just after clicking the Windows Start Button, search for `cmd` then open it.
- When the Window Terminal open, wait for the prompt to appear: `(C:\Users\You>)`.

[STEP 2] Upload `install-config-spboot-A1.env` & `setup-vps-spboot-A1.sh` files to your server:

- From Workbench Commands file, copy this line: `scp C:\TSS1\install-config-spboot-A1.env C:\TSS1\setup-vps-spboot-A1.sh root@your-server-ip:/root/` and paste it into the Windows CMD Terminal (**Caution:** Make sure you replaced `root@your-server-ip` with your details as instructed on **Section 3.3**) then click Enter to execute the command. The Terminal will ask for your **VPS password**, paste your password and click Enter.
- Wait for the Terminal to finish uploading until you see: `C:\Users\You>` then skip to [STEP 3].

While trying to run the command `scp` and you get an error similar to: “`scp' is not recognised`”, this means **OpenSSH** is missing on your Windows and you need to install it. To install it, choose options below:

1. **To install it by Windows feature:** On your PC click Windows Start Button > Settings > System > Optional Features > View Features > See Available Features > scroll & tick little box on: `OpenSSH.Client` > click `ADD Button` to download & install.

2. **To install it by commands:** start by opening the Windows PowerShell: On your PC click Windows Start Button > All > Terminal then Run as Administrator. Or just after clicking the Windows Start Button, search for PowerShell then right click it > Run as Administrator.

Copy this line: `Add-WindowsCapability -Online -Name OpenSSH.Client~~~~0.0.1.0` and paste it into the Windows PowerShell Terminal then click Enter to execute the command.

[STEP 3] If you are planning to deploy your Spring Boot JAR file, upload it to the server:

- From Workbench Commands file, copy this line: `scp C:\TSS1\myapp.jar root@your-server-ip:/root/` and paste it into the Windows CMD Terminal (**Caution:** Make sure you replaced: `myapp.jar` & `root@your-server-ip` with your details as instructed on **Section 3.3**) then click Enter to execute the command. The Terminal will ask for your **VPS password**, paste your password and click Enter.
- Wait for the Terminal to finish uploading then close the Windows Terminal because we are not going to need it any more.

4.2. Convert Uploaded Files to Unix line endings

Since your VPS is running on Ubuntu operating system, when we upload files from Windows operating system to Ubuntu operating system, some characters inside the files are still in Windows style and Ubuntu won't be able to read them. Now that we've uploaded our setup script and other supporting files, we have to convert two files to Unix style using command lines. For this we are going to use Putty Terminal.

[STEP 1] Open Putty Terminal

- If you had closed your Putty Terminal re-open it, (if it's offline) close it and re-open it. To open it, click Windows Start Button > All > Putty > Putty then open it. Or just after clicking the Windows Start Button, search for putty then open it.
- Since we've already configured Putty with your server credentials, to re-connect to your VPS server, we just have to load your saved settings then establish the connection.
- When Putty window opens (See image below first), under Saved Sessions: click your saved settings name: **VPSkey** > click Load Button > click Open Button to connect. The Putty Terminal will show up (Black Screen) asking you: Login as: Type `root` or your Root Username > then Click Enter.
- Wait for it to load your details until it displays something like this: `root@server:~#`, this means it's done processing and it's now ready to take your next command.

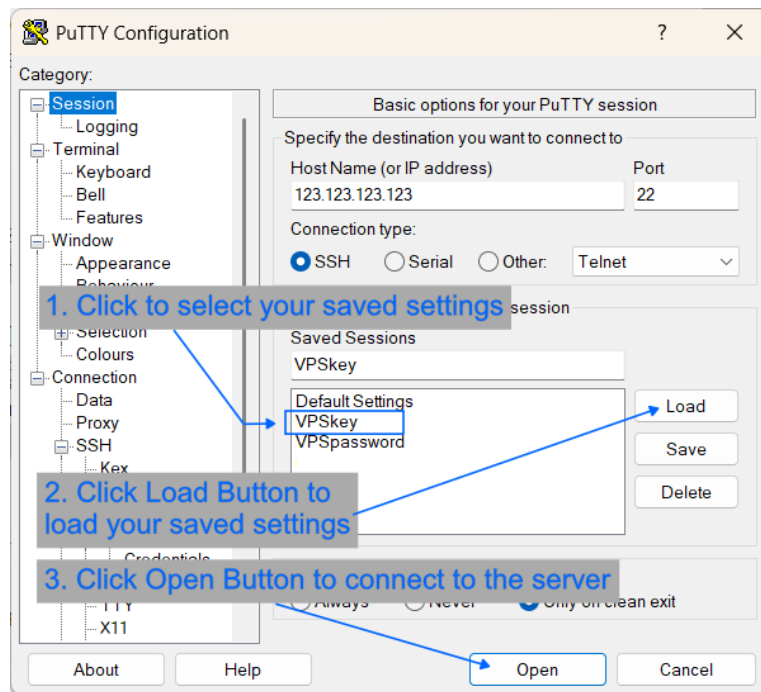


Figure 4.1: Reconnecting with Putty

[STEP 2] Convert two uploaded files to Unix style format:

- Copy this line: `sudo apt install dos2unix -y` and paste it into the Terminal cursor (You don't change anything on this command line) then click Enter to execute the command.
- Wait for the Terminal to finish processing until it displays something like this: `root@server:~#`.

Note: Sometimes the Terminal won't show you the process logs while processing, and when it's done processing, it will just show you this: `root@server:~#`, this means it's done processing and it's now ready to take your next command.

[STEP 3] Convert two uploaded files to Unix style format:

- Copy this line: `dos2unix install-config-spboot-A1.env setup-vps-spboot-A1.sh` and paste it into the Terminal cursor (You don't change anything on this command line) then click Enter to execute the command.
- Wait for the Terminal to finish processing until it displays something like this: `root@server:~#`, this means it's done processing and it's now ready to take your next command.

4.3. Grant Permission to Script

Now we need to tell the server to grant permission to the script so that it allows the script to run automatically.

[STEP 4] Grant permission to the script:

- Copy this line: `chmod +x setup-vps-spboot-A1.sh` and paste it into the Terminal cursor (You don't change anything on this command line) then click Enter to execute the command.
- Wait for the Terminal to finish processing until it displays something like this: `root@server:~#`, this means it's done processing and it's now ready to take your next command.

4.4. Install Screen & Start New Screen Session

This is a final step before we run our script. Now we need to open another special screen (Screen) that will allow us to see the script logs in progress and will also allow us to return to our script session logs when ever we get the network error that can cut us away from the script session. This means the new screen will allow the script to run in the background and keep the logs if we get the network error.

[STEP 5] Install Screen:

- Copy this line: `sudo apt update` and paste it into the Terminal cursor (You don't change anything on this command line) then click Enter to execute the command.
- Wait for the Terminal to finish processing until it displays something like this: `root@server:~#`, this means it's done processing and it's now ready to take your next command.
- Copy this line: `sudo apt install screen -y` and paste it into the Terminal cursor (You don't change anything on this command line) then click Enter to execute the command.
- Wait for the Terminal to finish processing until it displays something like this: `root@server:~#`, this means it's done processing and it's now ready to take your next command.

[STEP 6] Start new Screen session:

- Copy this line: `screen -S stack_setup` and paste it into the Terminal cursor (You don't change anything on this command line) then click Enter to execute the command. Then a new empty Screen session will appear on the Terminal.
- Wait for the Terminal to displays something like this: `root@root:~#` then proceed to the next step.

4.5. Run the Script

Now we are ready to run the server setup script.

[STEP 7] Run the Script:

- Copy this line: `sudo bash setup-vps-spboot-A1.sh` and paste it into the Terminal next to: `root@root:~#` (You don't change anything on this command line) then click Enter to execute the command then the script will start executing automatically and display the logs similar to the image below (See image below).

```
[INFO] Configuration loaded from: /root/install-config-spboot-A1.env
[SUCCESS] Configuration validation passed
[INFO] Domain:
[INFO] App Port: 8081
[INFO] Database: dummy (user: dummy, host: localhost:3306)
[INFO] Time Zone:
[INFO] Spring Profile: production

Spring Boot VPS Setup Script v3-A1 - Enhanced Edition

[INFO] Domain:
[INFO] App Port: 8081
[INFO] Database: dummy
[INFO] Time Zone:
[INFO] Log file: /var/log/springboot-setup.log

[STEP] Starting: Timezone Configuration
[SUCCESS] Timezone set to
[STEP] Starting: OS Update & Upgrade
[SUCCESS] Package lists updated
[SUCCESS] System upgraded
[STEP] Starting: Firewall (UFW)
[SUCCESS] Firewall configured and active
[STEP] Starting: Fail2Ban
[SUCCESS] Fail2Ban installed and running
[STEP] Starting: Java (OpenJDK 17)
[SUCCESS] Java installed: openjdk version "17.0.19" 2026-04-21
OpenJDK Runtime Environment (build 17.0.19+10-1-24.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 17.0.19+10-1-24.04.2-Ubuntu, mixed mode, sharing)
[STEP] Starting: MySQL
Synchronizing state of mysql.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable mysql
```

Figure 4.2: The script session logs

- During the runtime of the script, the script will handle the setup automatically, but keep on watching the Terminal logs. If ever any interactive prompt appears, just click **Enter** to proceed the script.

For example: While the script is upgrading the Operating System, The terminal might display an image similar to the one below (See image below first) and will ask you: “What do you want to do about modified configuration file sshd_config?”, You must select the option: “Keep local version currently installed” and click **Enter** for the script to proceed.

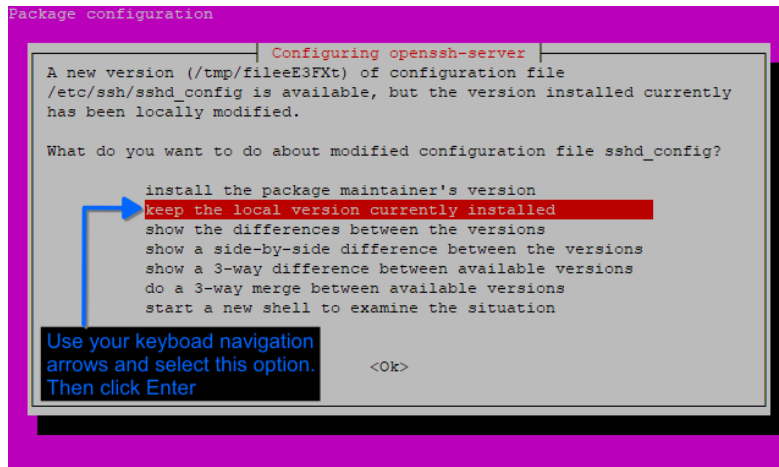


Figure 4.2: Dpkg prompt

- Just keep your eye on the Terminal logs until the script is finished, this may take a while about 10-20 minutes depending on your server and network properties.
- If for any reasons you find your Putty session loses connection from the server due to **time-outs error** or **connection error**, just close the Putty Terminal entirely and reopen it again. Then Re-establish the connection with your VPS, after you established the connection, you need to recall the Screen by running the command line: `screen -r stack_setup` then click **Enter** to return to the script session logs.
- When the script is done you will see the logs similar to the ones on the image below.

```
root@root: ~
=====
Installing Certbot and configuring SSL...
=====
SSL installation and auto-renewal configured.
[STEP] Starting: Deploy Spring Boot JAR
[INFO] Found myapp-springboot-1.0.0.jar in current directory
Created symlink /etc/systemd/system/multi-user.target.wants/springboot-app.service → /etc/systemd/system/springboot-app.service.
[SUCCESS] Spring Boot application deployed and running
[INFO] Created helper scripts: springboot-db-info, springboot-status

=====
                        SETUP COMPLETE!
=====

Installation Summary:
-----
✓ Timezone Configuration: Timezone
✓ OS Update & Upgrade: System updated and upgraded
✓ Firewall (UFW): Firewall active, ports 22,80,443 allowed
✓ Fail2Ban: Ban time: 3600s, Max retry: 5
✓ Java (OpenJDK 17): openjdk version "17.0.19" 2026-04-21
✓ MySQL: MySQL installed, DB: dummy
✓ Deploy Spring Boot JAR: JAR deployed with DB: dummy
-----
Success: 7 | Failed: 0 | Skipped: 0

📄 Configuration Summary:
Domain:
App Port: 8081
Database: dummy
DB User: dummy
DB Host: localhost:3306
Time Zone:
Spring Profile: production
JVM Heap: 256MB - 512MB

📋 NEXT STEPS:
2. Verify your application.properties matches these settings:
spring.datasource.url=jdbc:mysql://localhost:3306/dummy
spring.datasource.username=dummy
spring.datasource.password=dummy
server.port=8081

3. Useful commands:
springboot-status - Check application status
springboot-db-info - Show database configuration
sudo journalctl -u springboot-app -f - Follow logs
```

Figure 4.3: The script is finished

- When the script is done, you can screen-shot the Terminal logs for possible support.
- You can also get back to the install-config-spboot-A1 in TSS1 file and store the passwords in a safe place.
- Don't close Putty Terminal entirely, just close the Screen session that we just opened by running the command: `screen -X -S stack_setup quit` > then `click Enter`.

5. Testing & Verification

5.1. Testing your website if you deployed your JAR file.

- After the script is done with the setup, test your domain in browser/your custom client app.
- Check for SSL, in your browser address bar, check if your domain is starting with "https" and not "http".
- Browse through your website to ensure everything is running well including the database connection.
- If all pass, your VPS stack is live and you can now skip to `Section 7. Clean up`.
- If something is wrong, follow along `Section 5.2. and 6.Troubleshooting`.

5.2. Testing your website if you did not deploy your JAR file.

If you didn't deploy your JAR file, you cannot just test your website by entering your Domain into the browser and expect your browser to display something that's make sense. But if you do, you'll get a response: `502 Bad Gateway` or `Nginx default page` this means that Nginx has nothing to proxy to.

In this case you must test and verify your server setup installations through command lines and ensure by taking screen-shots of every Terminal response after running each command. You might as well note down these following list of Testing components from pierce of paper, then if one pass you just write pass next to it. This will help you identify those that failed and look them up on the Troubleshooting section. Below are the list of components you need to test and if all pass, your VPS stack is live.

5.2.1. Testing Nginx is running

- On Putty Terminal, run the command: `sudo systemctl status nginx` > then `click Enter`.

Pass if you see: Active: active (running) in green.

Fail if you see: Active: failed or inactive.

- If fail, run: `sudo journalctl -u nginx -n 50` > then `click Enter`. This will return errors, just screen-shot the errors and store it somewhere in your pc and follow the next testing (Save the screen-shot as: Testing Nginx is running).

5.2.2. Check if Nginx can respond to HTTP requests

- On Putty Terminal, run the command: `curl -I http://localhost` > then `click Enter`.

Pass if you see: `HTTP/1.1 200 OK` + `Server: nginx` or even `HTTP/1.1 502 Bad Gateway` is ok.

Fail if you see: `curl: (7) Fail to connect`.

- If fail by DNS problem, the problem is not nginx but the Domain propagation. Just screen-shot the errors and store it somewhere in your pc and follow the next testing (Save the screen-shot as: Check if Nginx can respond to HTTP requests).

5.2.3. Test domain point to VPS

- Open PC Terminal not Putty, run the command: `ping yourDomain.com` > then `click Enter`.

- Again run: `ping www.yourDomain.com` > then `click Enter`.

Pass if you see: IP in [] matches your VPS IP from `curl ifconfig.me`.

Fail if you see: Different IP.

- If fail by DNS problem, the problem is not nginx but the Domain propagation (must wait 2-24 hours). Just screen-shot the errors and store it somewhere in your pc and follow the next testing (Save the screen-shot as: Test domain point to VPS).

5.2.4. Test Firewall rules

- On Putty Terminal, run the command: `sudo ufw status verbose` > then `click Enter`.

Pass if you see: Nginx Full Allow Anywhere and 22/tcp Allow Anywhere.

Fail if you see: inactive.

- If fail, run: `sudo ufw enable` > then `click Enter`. Just screen-shot the errors and store it somewhere in your pc and follow the next testing (Save the screen-shot as: Test Firewall rules).

5.2.5. Test Nginx config syntax

- On Putty Terminal, run the command: `sudo nginx -t` > then click Enter.

Pass if you see: syntax is OK + test is successful.

Fail if you see: Shows line error.

- If fail: Just screen-shot the errors and store it somewhere in your pc and follow the next testing (Save the screen-shot as: Test Nginx config syntax).

If any fail, see Section **6. Troubleshooting**.

If all pass, jump to Section **7. Clean-up**.

6. Troubleshooting

If Section 5 failed, find your error here. 90% of issues are these ones.

6.1. Nginx won't start / systemctl status nginx shows failed

Symptom: Active: failed or inactive

Most common causes: Config syntax error or port 80 already in use.

Fix it:

1. Diagnose/Find out what's wrong: On Putty Terminal, run the command: `sudo nginx -t` > then click Enter. Analyse the last line of error and if you see error like: `port in use` / `already in use` > follow step 2. If you see error like: `syntax error line x` > follow step 3.

2. If "port in use" find what's in use & stop it:

- Let's find out what is using port 8080: On Putty Terminal, run the command: `sudo lsof -i :80` > then click Enter.

- Let's stop whatever is running on port 8080: On Putty Terminal run the command: `sudo kill -9 <PID>` > then click Enter.

- Let's restart Nginx on port 8080: On Putty Terminal run the command: `sudo systemctl restart nginx` > then click Enter. Screen-shot the Terminal response.

NOTES: If this were your only error from Section 5 Testing list, you should now go back to Section 5. Testing & Verification and perform the verification steps again to ensure that everything is well. Still stuck, contact our support desk coreligics@gmail.com.

3. If "syntax error" reinstall nginx: On Putty Terminal, open the nginx config file by running the command: `sudo nano /etc/nginx/sites-available/default` > then click Enter. Using your keyboard arrows, scroll down to find the error line and check the wrong syntax. Common nginx bugs for missing syntax elements are things like `“;”` at the end of the line or closing `“}”`. You can also compare the syntax error with the official example so that you can identify what's wrong or what's missing: open your browser: https://nginx.org/en/docs/beginners_guide.html scroll to server and compare.

To save Nginx config file after edits: On your keyboard, click `ctrl+o` to save > then click `Enter` > then click `ctrl+x` to exit the Nano Screen.

- Run the command: `sudo systemctl restart nginx` > then click Enter.

NOTES: If this were your only error from Section 5 Testing list, you should now go back to Section 5. Testing & Verification and perform the verification steps again to ensure that everything is well. Still stuck, contact our support desk coreligics@gmail.com.

7. Clean-up (Optional)

After the script is done and testing pass, you must keep your VPS clean and secure.

7.1. Clear apt Cache

Free disc space by removing old packages + debris files. This can save 200-500MB on your server.

- On Putty run this command line: `sudo apt autoremove -y` > then click Enter to execute.
- Run: `sudo apt autoclean` > then click Enter to execute.

7.2. Remove Script Files

You may want to free more disc space by removing our script files and your jar file that we uploaded.

- On Putty run this command line: `rm install-config-spboot-A1.env` > then click Enter to execute then wait for `root@server:~#`.
- Run: `rm setup-vps-spboot-A1.sh` > then click Enter to execute then wait for `root@server:~#`.
- If you deployed your JAR file. Run: `rm myapp.jar` (replace myapp.jar with your JAR file name) > then click Enter to execute then wait for `root@server:~#`.

7.3. Disable Password Authentication

You may need to disable password authentication for extra security on accessing your VPS back-end. Since you already configured the SSH Public key on your VPS server, we recommend to disable password authentication because password login is less secure than SSH keys. And since your PC is the only one that has the SSH Private key, it must be the only one to connect to your VPS back-end through SSH Keys authentication unless if you want to configure other own PCs with your Private key.

This can stop hackers from brute-forcing your password because the key system is the hardest to be guessed by bots than the password.

Caution: Only do this after you confirmed you SSH key login works 100%. **Don't do this if you're not sure.**

[Step 1] Open SSH Config File

- On Putty run this command line: `nano /etc/ssh/sshd_config` > then click Enter to execute then wait for the Nano Text editor to display a `sshd_config` file.

[Step 2] Find a Variable in a File

- In Nano, use your keyboard navigation arrows to move the cursor and scroll down to find this line: `#PasswordAuthentication yes` then change the value from `yes` to `no` and also remove the symbol `#`. The results should look exactly like this: `PasswordAuthentication no`.

[Step 3] Save & Restart SSH

- On your keyboard: Click `ctrl+o` to save changes > then click Enter > click `ctrl+x` to exit Nano.
- To restart SSH, on Putty run this command line: `systemctl restart ssh` > then click Enter to restart.

8. Feedback & Support

If this script saved you time, please leave your feedback where you purchased the script. It takes 30 seconds and helps other developers find it.

Found a bug or have any question?

1. First check Section 7 for Troubleshooting above
2. Still stuck? Email us: coreligics@gmail.com

Leave a review

1. Bought on Gumroad? Go to “Leave Review” on your download page.
2. Bought on Etsy? Go to “Orders” > “Leave Review”.

Your feedback = We improve the script faster for you...

9. Next Steps & Best Practices

Database management

For managing your Database securely, we recommend to use GUI tools.

Recommended Tools

- MySQL Workbench (Free): Windows/Mac/Linux. Best for MySQL/MariaDB.
- DBeaver (Free): Works with MySQL, Postgres, etc.
- phpMyAdmin (Free): Web panel. Install it on your server.

Final Notes: This manual ends here...

Spring Boot VPS Deployment Guide v1.0

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